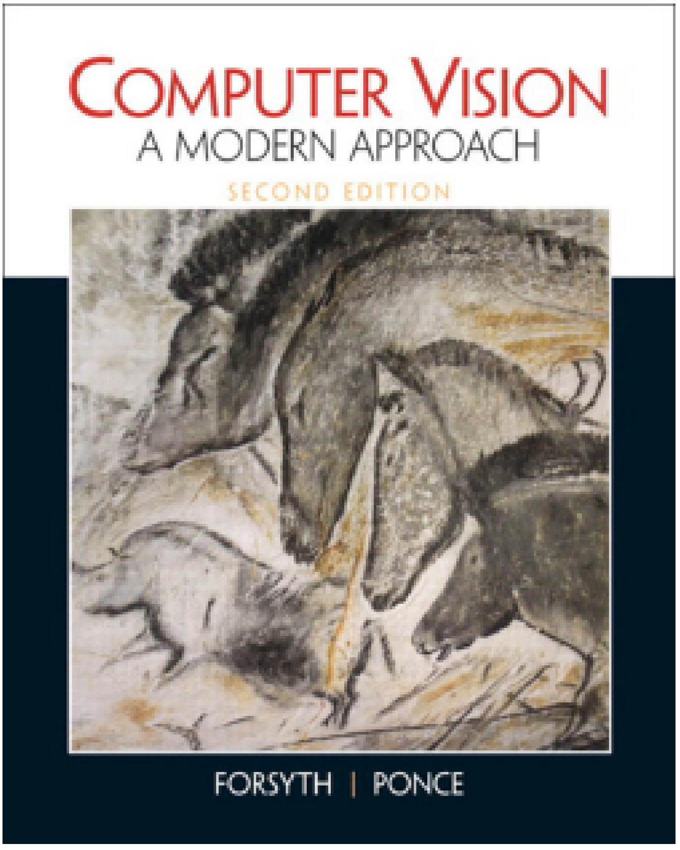

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COMPUTER VISION A MODERN APPROACH

second edition

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To my family—DAF

To my father, Jean-Jacques Ponce —JP

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Contents

II EARLY VISION: JUST ONE IMAGE 105

VI APPLICATIONS AND TOPICS 557

VII BACKGROUND MATERIAL 661

Preface

Computer vision as a field is an intellectual frontier. Like any frontier, it is exciting and disorganized, and there is often no reliable authority to appeal to. Many useful ideas have no theoretical grounding, and some theories are useless in practice; developed areas are widely scattered, and often one looks completely inaccessible from the other. Nevertheless, we have attempted in this book to present a fairly orderly picture of the field.

We see computer vision—or just "vision"; apologies to those who study human or animal vision—as an enterprise that uses statistical methods to disentangle data using models constructed with the aid of geometry, physics, and learning theory. Thus, in our view, vision relies on a solid understanding of cameras and of the physical process of image formation (Part I of this book) to obtain simple inferences from individual pixel values (Part II), combine the information available in multiple images into a coherent whole (Part III), impose some order on groups of pixels to separate them from each other or infer shape information (Part IV), and recognize objects using geometric information or probabilistic techniques (Part V). Computer vision has a wide variety of applications, both old (e.g., mobile robot navigation, industrial inspection, and military intelligence) and new (e.g., human computer interaction, image retrieval in digital libraries, medical image analysis, and the realistic rendering of synthetic scenes in computer graphics). We discuss some of these applications in part VII.

IN THE SECOND EDITION

We have made a variety of changes since the first edition, which we hope have improved the usefulness of this book. Perhaps the most important change follows a big change in the discipline since the last edition. Code and data are now widely published over the Internet. It is now quite usual to build systems out of other people's published code, at least in the first instance, and to evaluate them on other people's datasets. In the chapters, we have provided guides to experimental resources available online. As is the nature of the Internet, not all of these URL's will work all the time; we have tried to give enough information so that searching Google with the authors' names or the name of the dataset or codes will get the right result.

Other changes include:

- We have simplified. We give a simpler, clearer treatment of mathematical topics. We have particularly simplified our treatment of cameras (Chapter 1), shading (Chapter 2), and reconstruction from two views (Chapter 7) and from multiple views (Chapter 8)
- We describe a broad range of applications, including image-based mod elling and rendering (Chapter 19), image search (Chapter 22), building image mosaics (Section 12.1), medical image registration (Section 12.3), interpreting range data (Chapter 14), and understanding human activity (Chapter 21).

- We have written a comprehensive treatment of the modern features, par ticularly HOG and SIFT (both in Chapter 5), that drive applications ranging from building image mosaics to object recognition.
- We give a detailed treatment of modern image editing techniques, in cluding removing shadows (Section 3.5), filling holes in images (Section 6.3), noise removal (Section 6.4), and interactive image segmentation (Section 9.2).
- We give a comprehensive treatment of modern object recognition tech niques. We start with a practical discussion of classifiers (Chapter 15); we then describe standard methods for image classification techniques (Chapter 16), and object detection (Chapter 17). Finally, Chapter 18 reviews a wide range of recent topics in object recognition.
- Finally, this book has a very detailed index, and a bibliography that is as comprehensive and up-to-date as we could make it.

WHY STUDY VISION?

Computer vision's great trick is extracting descriptions of the world from pictures or sequences of pictures. This is unequivocally useful. Taking pictures is usually nondestructive and sometimes discreet. It is also easy and (now) cheap. The de scriptions that users seek can differ widely between applications. For example, a technique known as structure from motion makes it possible to extract a representa tion of what is depicted and how the camera moved from a series of pictures. People in the entertainment industry use these techniques to build three-dimensional (3D) computer models of buildings, typically keeping the structure and throwing away the motion. These models are used where real buildings cannot be; they are set fire to, blown up, etc. Good, simple, accurate, and convincing models can be built from quite small sets of photographs. People who wish to control mobile robots usually keep the motion and throw away the structure. This is because they generally know something about the area where the robot is working, but usually don't know the precise robot location in that area. They can determine it from information about how a camera bolted to the robot is moving.

There are a number of other, important applications of computer vision. One is in medical imaging: one builds software systems that can enhance imagery, or identify important phenomena or events, or visualize information obtained by imag ing. Another is in inspection: one takes pictures of objects to determine whether they are within specification. A third is in interpreting satellite images, both for military purposes (a program might be required to determine what militarily inter esting phenomena have occurred in a given region recently; or what damage was caused by a bombing) and for civilian purposes (what will this year's maize crop be? How much rainforest is left?) A fourth is in organizing and structuring collec tions of pictures. We know how to search and browse text libraries (though this is a subject that still has difficult open questions) but don't really know what to do with image or video libraries.

Computer vision is at an extraordinary point in its development. The subject itself has been around since the 1960s, but only recently has it been possible to build useful computer systems using ideas from computer vision. This flourishing

has been driven by several trends: Computers and imaging systems have become very cheap. Not all that long ago, it took tens of thousands of dollars to get good digital color images; now it takes a few hundred at most. Not all that long ago, a color printer was something one found in few, if any, research labs; now they are in many homes. This means it is easier to do research. It also means that there are many people with problems to which the methods of computer vision apply. For example, people would like to organize their collections of photographs, make 3D models of the world around them, and manage and edit collections of videos. Our understanding of the basic geometry and physics underlying vision and, more important, what to do about it, has improved significantly. We are beginning to be able to solve problems that lots of people care about, but none of the hard problems have been solved, and there are plenty of easy ones that have not been solved either (to keep one intellectually fit while trying to solve hard problems). It is a great time to be studying this subject.

What Is in this Book

This book covers what we feel a computer vision professional ought to know. How ever, it is addressed to a wider audience. We hope that those engaged in compu tational geometry, computer graphics, image processing, imaging in general, and robotics will find it an informative reference. We have tried to make the book accessible to senior undergraduates or graduate students with a passing interest in vision. Each chapter covers a different part of the subject, and, as a glance at Table 1 will confirm, chapters are relatively independent. This means that one can dip into the book as well as read it from cover to cover. Generally, we have tried to make chapters run from easy material at the start to more arcane matters at the end. Each chapter has brief notes at the end, containing historical material and assorted opinions. We have tried to produce a book that describes ideas that are useful, or likely to be so in the future. We have put emphasis on understanding the basic geometry and physics of imaging, but have tried to link this with actual ap plications. In general, this book reflects the enormous recent influence of geometry and various forms of applied statistics on computer vision.

Reading this Book

A reader who goes from cover to cover will hopefully be well informed, if exhausted; there is too much in this book to cover in a one-semester class. Of course, prospec tive (or active) computer vision professionals should read every word, do all the exercises, and report any bugs found for the third edition (of which it is probably a good idea to plan on buying a copy!). Although the study of computer vision does not require deep mathematics, it does require facility with a lot of different math ematical ideas. We have tried to make the book self-contained, in the sense that readers with the level of mathematical sophistication of an engineering senior should be comfortable with the material of the book and should not need to refer to other texts. We have also tried to keep the mathematics to the necessary minimum—after all, this book is about computer vision, not applied mathematics—and have chosen to insert what mathematics we have kept in the main chapter bodies instead of a separate appendix.

TABLE 1: Dependencies between chapters: It will be difficult to read a chapter if you don’t have a good grasp of the material in the chapters it “requires.” If you have not read the chapters labeled “helpful,” you might need to look up one or two things.

Part	Chapter	Requires	Helpful
I	1: Geometric Camera Models		
	2: Light and Shading		
	3: Color	2	

II 4: Linear Filters 5: Local Image Features 4
6: Texture 5, 4 2

III 7: Stereopsis 1 22 8: Structure from Motion 1, 7 22

IV 9: Segmentation by Clustering 2, 3, 4, 5, 6, 22 10: Grouping and Model Fitting 9 11: Tracking 2, 5, 22

V 12: Registration 1 14 13: Smooth Surfaces and Their Outlines 1
14: Range Data 12 15: Learning to Classify 22 16: Classifying Images 15, 5
17: Detecting Objects in Images 16, 15, 5
18: Topics in Object Recognition 17, 16, 15, 5

VI 19: Image-Based Modeling and Rendering 1, 2, 7, 8 20: Looking at People 17, 16, 15, 11, 5 21: Image Search and Retrieval 17, 16, 15, 11, 5

VII 22: Optimization Techniques

Generally, we have tried to reduce the interdependence between chapters, so that readers interested in particular topics can avoid wading through the whole book. It is not possible to make each chapter entirely self-contained, however, and Table 1 indicates the dependencies between chapters.

We have tried to make the index comprehensive, so that if you encounter a new term, you are likely to find it in the book by looking it up in the index. Computer vision is now fortunate in having a rich range of intellectual resources. Software and datasets are widely shared, and we have given pointers to useful datasets and software in relevant chapters; you can also look in the index, under “software” and under “datasets,” or under the general topic.

We have tried to make the bibliography comprehensive, without being over whelming. However, we have not been able to give complete bibliographic references for any topic, because the literature is so large.

What Is Not in this Book

The computer vision literature is vast, and it was not easy to produce a book about computer vision that could be lifted by ordinary mortals. To do so, we had to cut material, ignore topics, and so on.

We left out some topics because of personal taste, or because we became exhausted and stopped writing about a particular area, or because we learned about them too late to put them in, or because we had to shorten some chapter, or because we didn’t understand them, or any of hundreds of other reasons. We have tended to omit detailed discussions of material that is mainly of historical interest, and offer instead some historical remarks at the end of each chapter.

We have tried to be both generous and careful in attributing ideas, but neither of us claims to be a fluent intellectual archaeologist, and computer vision is a very big topic indeed. This means that some ideas may have deeper histories than we have indicated, and that we may have omitted citations.

There are several recent textbooks on computer vision. Szeliski (2010) deals with the whole of vision. Parker (2010) deals specifically with algorithms. Davies (2005) and Steger et al. (2008) deal with practical applications, particularly regis tration. Bradski and Kaehler (2008) is an introduction to

OpenCV, an important open-source package of computer vision routines.

There are numerous more specialized references. Hartley and Zisserman (2000a) is a comprehensive account of what is known about multiple view ge ometry and estimation of multiple view parameters. Ma et al. (2003b) deals with 3D reconstruction methods. Cyganek and Siebert (2009) covers 3D reconstruction and matching. Paragios et al. (2010) deals with mathematical models in computer vision. Blake et al. (2011) is a recent summary of what is known about Markov random field models in computer vision. Li and Jain (2005) is a comprehensive account of face recognition. Moeslund et al. (2011), which is in press at time of writing, promises to be a comprehensive account of computer vision methods for watching people. Dickinson et al. (2009) is a collection of recent summaries of the state of the art in object recognition. Radke (2012) is a forthcoming account of computer vision methods applied to special effects.

Much of computer vision literature appears in the proceedings of various con ferences. The three main conferences are: the IEEE Conference on Computer Vision and Pattern Recognition (CVPR); the IEEE International Conference on Computer Vision (ICCV); and the European Conference on Computer Vision. A significant fraction of the literature appears in regional conferences, particularly the Asian Conference on Computer Vision (ACCV) and the British Machine Vi sion Conference (BMVC). A high percentage of published papers are available on the web, and can be found with search engines; while some papers are confined to pay-libraries, to which many universities provide access, most can be found without cost.

ACKNOWLEDGMENTS

In preparing this book, we have accumulated a significant set of debts. A number of anonymous reviewers read several drafts of the book for both first and second edition and made extremely helpful contributions. We are grateful to them for their time and efforts.

Our editor for the first edition, Alan Apt, organized these reviews with the

help of Jake Warde. We thank them both. Leslie Galen, Joe Albrecht, and Dianne Parish, of Integre Technical Publishing, helped us overcome numerous issues with proofreading and illustrations in the first edition.

Our editor for the second edition, Tracy Dunkelberger, organized reviews with the help of Carole Snyder. We thank them both. We thank Marilyn Lloyd for helping us get over various production problems.

Both the overall coverage of topics and several chapters were reviewed by various colleagues, who made valuable and detailed suggestions for their revision. We thank Narendra Ahuja, Francis Bach, Kobus Barnard, Margaret Fleck, Martial Hebert, Julia Hockenmaier, Derek Hoiem, David Kriegman, Jitendra Malik, and Andrew Zisserman.

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We have been very lucky to have colleagues at various universities use (of ten rough) drafts of our book in their vision classes. Institutions whose students suffered through these drafts include, in addition to ours, Carnegie-Mellon Univer sity, Stanford University, the University of Wisconsin at Madison, the University of California at Santa Barbara and the University of Southern California; there may be others we are not aware of. We are grateful for all the helpful comments from adopters, in particular Chris Bregler, Chuck Dyer, Martial Hebert, David Krieg man, B.S. Manjunath, and Ram Nevatia, who sent us many detailed and helpful comments and corrections.

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In the first edition, we said

> If you find an apparent typographic error, please email DAF... with the details, using the phrase “book typo” in your email; we will try to credit the first finder of each typo in the second edition.

which turns out to have been a mistake. DAF’s ability to manage and preserve

email logs was just not up to this challenge. We thank all finders of typographic errors; we have tried to fix the errors and have made efforts to credit all the people who have helped us.

We also thank P. Besl, B. Boufama, J. Costeira, P. Debevec, O. Faugeras, Y. Genc, M. Hebert, D. Huber, K. Ikeuchi, A.E. Johnson, T. Kanade, K. Kutulakos, M. Levoy, Y. LeCun, S. Mahamud, R. Mohr, H. Moravec, H. Murase, Y. Ohta, M. Okutami, M. Pollefeys, H. Saito, C. Schmid, J. Shotton, S. Sullivan, C. Tomasi, and M. Turk for providing the originals of some of the figures shown in this book.

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Figures: Some images used herein were obtained from IMSI’s Master Photos Collection, 1895 Francisco Blvd. East, San Rafael, CA 94901-5506, USA. We have made extensive use of figures from the published literature; these figures are credited in their captions. We thank the copyright holders for

extending permission to use these figures.

Bibliography: In preparing the bibliography, we have made extensive use of Keith Price's excellent computer vision bibliography, which can be found at http://iris.usc.edu/Vision-Notes/bibliography/contents.html.

TABLE 2: A one-semester introductory class in computer vision for seniors or first-year graduate students in computer science, electrical engineering, or other engineering or science disciplines.

Week	Chapter	Sections	Key topics
1	1, 2	1.1, 2.1, 2.2.x	pinhole cameras, pixel shading models, one inference from shading example
2	3	3.1–3.5	human color perception, color physics, color spaces, image color model
3	4	all	linear filters
4	5	all	building local features
5	6	6.1, 6.2	texture representations from filters, from vector quantization
6	7	7.1, 7.2	binocular geometry, stereopsis
7	8	8.1	structure from motion with perspective cameras
8	9	9.1–9.3	segmentation ideas, applications, segmentation by clustering pixels
9	10	10.1–10.4	Hough transform, fitting lines, robustness, RANSAC,
10	11	11.1-11.3	simple tracking strategies, tracking by matching, Kalman filters, data association
11	12	all	registration
12	15	all	classification
13	16	all	classifying images
14	17	all	detection
15	choice	all	one of chapters 14, 19, 20, 21 (application topics)

SAMPLE SYLLABUSES

The whole book can be covered in two (rather intense) semesters, by starting at the first page and plunging on. Ideally, one would cover one application chapter—probably the chapter on image-based rendering—in the first semester, and the other one in the second. Few departments will experience heavy demand for such a de tailed sequence of courses. We have tried to structure this book so that instructors can choose areas according to taste. Sample syllabuses for busy 15-week semesters appear in Tables 2 to 6, structured according to needs that can reasonably be ex pected. We would encourage (and expect!) instructors to rearrange these according to taste.

Table 2 contains a suggested syllabus for a one-semester introductory class in computer vision for seniors or first-year graduate students in computer science, electrical engineering, or other engineering or science disciplines. The students receive a broad presentation of the field, including application areas such as digital libraries and image-based rendering. Although the hardest theoretical material is omitted, there is a thorough treatment of the basic geometry and physics of image formation. We assume that students will have a wide range of backgrounds, and can be assigned background readings in probability. We have put off the application chapters to the end, but many may prefer to cover them earlier.

Table 3 contains a syllabus for students of computer graphics who want to know the elements of vision that are relevant to their topic. We have emphasized methods that make it possible to recover object models from image information;

TABLE 3: A syllabus for students of computer graphics who want to know the elements of vision that are relevant to their topic.

Week	Chapter	Sections	Key topics
1	1, 2	1.1, 2.1, 2.2.4	pinhole cameras, pixel shading models, photometric stereo
2	3	3.1–3.5	human color perception, color physics, color spaces, image color model
3	4	all	linear filters
4	5	all	building local features
5	6	6.3, 6.4	texture synthesis, image denoising
6	7	7.1, 7.2	binocular geometry, stereopsis
7	7	7.4, 7.5	advanced stereo methods
8	8	8.1	structure from motion with perspective cameras
9	10	10.1–10.4	Hough transform, fitting lines, robustness, RANSAC,
10	9	9.1–9.3	segmentation ideas, applications, segmentation by clustering pixels
11	11	11.1-11.3	simple tracking strategies, tracking by matching, Kalman filters, data association
12	12	all	registration
13	14	all	range data
14	19	all	image-based modeling and rendering
15	13	all	surfaces and outlines

understanding these topics needs a working knowledge of cameras and filters. Track ing is becoming useful in the graphics world, where it is particularly important for motion capture. We assume that students will have a wide range of backgrounds, and have some exposure to probability.

Table 4 shows a syllabus for students who are primarily interested in the applications of computer vision. We cover material of most immediate practical interest. We assume that students will have a wide range of backgrounds, and can be assigned background reading.

Table 5 is a suggested syllabus for students of cognitive science or artificial intelligence who want a basic outline of the important notions of computer vision. This syllabus is less aggressively paced, and assumes less mathematical experience.

Our experience of teaching computer vision is that no single idea presents any particular conceptual difficulties, though some are harder than others. Difficulties are caused by the tremendous number of new ideas required by the subject. Each subproblem seems to require its own way of thinking, and new tools to cope with it. This makes learning the subject rather daunting. Table 6 shows a sample syllabus for students who are really not bothered by these difficulties. They would need to have quite a strong interest in applied mathematics, electrical engineering or physics, and be very good at picking things up as they go along. This syllabus sets a furious pace, and assumes that students can cope with a lot of new material.

NOTATION

We use the following notation throughout the book: Points, lines, and planes are denoted by Roman or Greek letters in italic font (e.g., P, Δ, or Π). Vectors are

TABLE 4: A syllabus for students who are primarily interested in the applications of computer vision.

Week Chapter Sections Key topics

1 1, 2 1.1, 2.1, 2.2.4 pinhole cameras, pixel shading models, photometric stereo
2 3 3.1–3.5 human color perception, color physics, color spaces, image color model
3 4 all linear filters
4 5 all building local features
5 6 6.3, 6.4 texture synthesis, image denoising 6 7 7.1, 7.2 binocular geometry, stereopsis
7 7 7.4, 7.5 advanced stereo methods
8 8, 9 8.1, 9.1–9.2 structure from motion with perspective cameras, segmentation ideas, applications
9 10 10.1–10.4 Hough transform, fitting lines, robustness, RANSAC, 10 12 all registration
11 14 all range data
12 16 all classifying images
13 19 all image based modeling and rendering 14 20 all looking at people
15 21 all image search and retrieval

usually denoted by Roman or Greek bold-italic letters (e.g., v, P , or ξ), but the vector joining two points P and Q is often denoted by $\overrightarrow{PQ}$. Lower-case letters are normally used to denote geometric figures in the image plane (e.g., p, p, δ), and upper-case letters are used for scene objects (e.g., P, Π). Matrices are denoted by Roman letters in calligraphic font (e.g., U).

The familiar three-dimensional Euclidean space is denoted by E^3, and the vector space formed by n-tuples of real numbers with the usual laws of addition and multiplication by a scalar is denoted by R^n, with 0 being used to denote the zero vector. Likewise, the vector space formed by m × n matrices with real entries is denoted by $R^{m\times n}$. When m = n, Id is used to denote the identity matrix—that is, the n × n matrix whose diagonal entries are equal to 1 and nondiagonal entries are equal to 0. The transpose of the m × n matrix U with coefficients u_{ij} is the n × m matrix denoted by U^T with coefficients u_{ji}. Elements of R^n are often identified with column vectors or n × 1 matrices, for example, $a = (a_1, a_2, a_3)^T$ is the transpose of a 1 × 3 matrix (or row vector), i.e., an 3 × 1 matrix (or column vector), or equivalently an element of R^3.

The dot product (or inner product) of two vectors $a = (a_1, \ldots, a_n)^T$ and $b = (b_1, \ldots, b_n)^T$ in R^n is defined by

$$a \cdot b = a_1 b_1 + \cdots + a_n b_n,$$

and it can also be written as a matrix product, i.e., $a \cdot b = a^T b = b^T a$. We denote by $|a|^2 = a \cdot a$ the square of the Euclidean norm of the vector a and denote by d the distance function induced by the Euclidean norm in E^n, i.e., $d(P, Q) = |\overrightarrow{PQ}|$. Given a matrix U in $R^{m\times n}$, we generally use |U| to denote its Frobenius norm, i.e., the square root of the sum of its squared entries.

TABLE 5: For students of cognitive science or artificial intelligence who want a basic outline of the important notions of computer vision.

Week Chapter Sections Key topics

1 1, 2 1.1, 2.1, 2.2.x pinhole cameras, pixel shading models, one inference from shading example
2 3 3.1–3.5 human color perception, color physics, color spaces, image color model
3 4 all linear filters
4 5 all building local features
5 6 6.1, 6.2 texture representations from filters, from vector quantization
6 7 7.1, 7.2 binocular geometry, stereopsis
8 9 9.1–9.3 segmentation ideas, applications, segmentation by clustering pixels
9 11 11.1, 11.2 simple tracking strategies, tracking using matching, optical flow
10 15 all classification
11 16 all classifying images
12 20 all looking at people
13 21 all image search and retrieval
14 17 all detection
15 18 all topics in object recognition

When the vector a has unit norm, the dot product a·b is equal to the (signed) length of the projection of b onto a. More generally,

$$a \cdot b = |a||b| \cos \theta,$$

where θ is the angle between the two vectors, which shows that a necessary and sufficient condition for two vectors to be orthogonal is that their dot product be zero.

The cross product (or outer product) of two vectors $a = (a_1, a_2, a_3)^T$ and $b = (b_1, b_2, b_3)^T$ in R^3 is the vector

$$a \times b \stackrel{def}{=} \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}.$$

Note that $a \times b = [a_\times]b$, where

$$[a_\times] \stackrel{def}{=} \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}.$$

The cross product of two vectors a and b in R^3 is orthogonal to these two vectors, and a necessary and sufficient condition for a and b to have the same direction is that $a \times b = 0$. If θ denotes as before the angle between the vectors a and b, it can be shown that

$$|a \times b| = |a||b||\sin \theta|.$$

TABLE 6: A syllabus for students who have a strong interest in applied mathematics, electrical engineering, or physics.

Week Chapter Sections Key topics

1 1, 2 all; 2.1–2.4 cameras, shading 2 3 all color

3 4 all linear filters
4 5 all building local features
5 6 all texture
6 7 all stereopsis
7 8 all structure from motion with perspective cameras 8 9 all segmentation by clustering pixels 9 10 all fitting models
10 11 11.1–11.3 simple tracking strategies, tracking by matching, Kalman filters, data association
11 12 all registration
12 15 all classification
13 16 all classifying images
14 17 all detection
15 choice all one of chapters 14, 19, 20, 21

PROGRAMMING ASSIGNMENTS AND RESOURCES

The programming assignments given throughout this book sometimes require rou tines for numerical linear algebra, singular value decomposition, and linear and nonlinear least squares. An extensive set of such routines is available in MATLAB as well as in public-domain libraries such as LINPACK, LAPACK, and MINPACK, which can be downloaded from the Netlib repository (http://www.netlib.org/). In the text, we offer extensive pointers to software published on the Web and to datasets published on the Web. OpenCV is an important open-source package of computer vision routines (see Bradski and Kaehler (2008)).

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PART ONE

IMAGE FORMATION

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CHAPTER 1

Geometric Camera Models

There are many types of imaging devices, from animal eyes to video cameras and radio telescopes, and they may or may not be equipped with lenses. For example, the first models of the camera obscura (literally, dark chamber) invented in the sixteenth century did not have lenses, but instead used a pinhole to focus light rays onto a wall or translucent plate and demonstrate the laws of perspective discovered a century earlier by Brunelleschi. Pinholes were replaced by more and more sophis ticated lenses as early as 1550, and the modern photographic or digital camera is essentially a camera obscura capable of recording the amount of light striking every small area of its backplane (Figure 1.1).

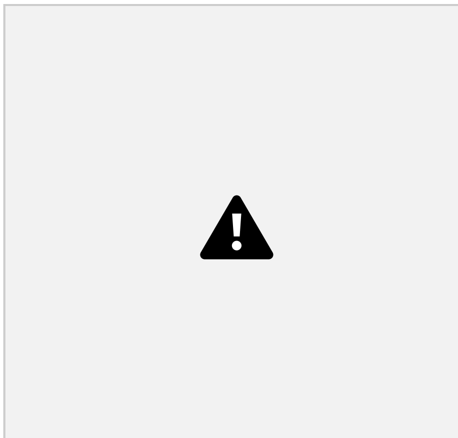


FIGURE 1.1: Image formation on the backplate of a photographic camera. Figure from US NAVY MANUAL OF BASIC OPTICS AND OPTICAL INSTRUMENTS, prepared by the Bureau of Naval Personnel, reprinted by Dover Publications, Inc. (1969).

The imaging surface of a camera is in general a rectangle, but the shape of the human retina is much closer to a spherical surface, and panoramic cameras may be equipped with cylindrical retinas. Imaging sensors have other characteristics. They may record a spatially discrete picture (like our eyes with their rods and cones, 35mm cameras with their grain, and digital cameras with their rectangular picture elements, or pixels), or a continuous one (in the case of old-fashioned TV tubes, for example). The signal that an imaging sensor records at a point on its retina may itself be discrete or continuous, and it may consist of a single number (as for a black-and-white camera), a few values (e.g., the RGB intensities for a color camera or the responses of the three types of cones for the human eye), many numbers (e.g., the responses of hyperspectral sensors), or even a continuous function of wavelength (which is essentially the case for spectrometers). Chapter 2

considers cameras as radiometric devices for measuring light energy, brightness, and color. Here, we focus instead on purely geometric camera characteristics. After introducing several models of image formation in Section 1.1—including a brief description of this process in the human eye in Section 1.1.4—we define the intrinsic and extrinsic geometric parameters characterizing a camera in Section

1.2, and finally show how to estimate these parameters from image data—a process known as geometric camera calibration—in Section 1.3.

1.1 IMAGE FORMATION

1.1.1 Pinhole Perspective

Imagine taking a box, using a pin to prick a small hole in the center of one of its sides, and then replacing the opposite side with a translucent plate. If you hold that box in front of you in a dimly lit room, with the pinhole facing some light source, say a candle, an inverted image of the candle will appear on the translucent plate (Figure 1.2). This image is formed by light rays issued from the scene facing the box. If the pinhole were really reduced to a point (which is physically impossible, of course), exactly one light ray would pass through each point in the plane of the plate (or image plane), the pinhole, and some scene point.

image
plane

I
imag
e

pinhole

virtua

FIGURE 1.2: The pinhole imaging model.

In reality, the pinhole will have a finite (albeit small) size, and each point in the image plane will collect light from a cone of rays subtending a finite solid angle, so this idealized and extremely simple model of the imaging geometry will not strictly apply. In addition, real cameras are normally equipped with lenses, which further complicates things. Still, the pinhole perspective (also called central perspective) projection model, first proposed by Brunelleschi at the beginning of the fifteenth century, is mathematically convenient and, despite its simplicity, it often provides an acceptable approximation of the imaging process. Perspective projection creates inverted images, and it is sometimes convenient to consider instead a virtual image associated with a plane lying in front of the pinhole, at the same distance from it as the actual image plane (Figure 1.2). This virtual image is not inverted but is otherwise strictly equivalent to the actual one. Depending on the context, it may be more convenient to think about one or the other. Figure 1.3 (a) illustrates an obvious effect of perspective projection: the apparent size of objects depends on their distance. For example, the images b and c of the posts B and C have the same height, but A and C are really half the size of B. Figure 1.3 (b) illustrates

another well-known effect: the projections of two parallel lines lying in some plane Φ appear to converge on a horizon line h formed by the intersection of the image plane Π with the plane parallel to Φ and passing through the pinhole. Note that the line L parallel to Π in Φ has no image at all.

B

C a
O
b
c O

A

(a) d d d Π

h
L
Π

(b) Φ

FIGURE 1.3: Perspective effects: (a) far objects appear smaller than close ones: The distance d from the pinhole O to the plane containing C is half the distance from O to the plane containing A and B; (b) the images of parallel lines intersect at the horizon (after Hilbert and Cohn-Vossen, 1952, Figure 127). Note that the image plane Π is behind the pinhole in (a) (physical retina), and in front of it in (b) (virtual image plane). Most of the diagrams in this chapter and in the rest of this book will feature the physical image plane, but a virtual one will also be used when appropriate, as in (b).

These properties are easy to prove in a purely geometric fashion. As usual, however, it is often convenient (if not quite as elegant) to reason in terms of reference frames, coordinates, and equations. Consider, for example, a coordinate system (O, i, j, k) attached to a pinhole camera, whose origin O coincides with the pinhole, and vectors i and j form a basis for a vector plane parallel to the image plane Π, itself located at a positive distance d from the pinhole along the vector k (Figure 1.4). The line perpendicular to Π and passing through the pinhole is called the optical axis, and the point c where it pierces Π is called the image center. This point can be used as the origin of an image plane coordinate frame, and it plays an important role in camera calibration procedures.

Let P denote a scene point with coordinates (X, Y, Z) and p denote its image

k **i**

P

c

p

j Π d

O

FIGURE 1.4: The perspective projection equations are derived in this section from the collinearity of the point P, its image p, and the pinhole O.

with coordinates (x, y, z). (Throughout this chapter, we will use uppercase letters to denotes points in space, and lowercase letters to denote their image projections.) Since p lies in the image plane, we have z = d. Since the three points P, O, and p are collinear, we have $\overrightarrow{Op} = \lambda \overrightarrow{OP}$ for some number λ, so

$$\begin{cases} x = \lambda X \\ y = \lambda Y \\ d = \lambda Z \end{cases} \iff \lambda = \frac{x}{X} = \frac{y}{Y} = \frac{d}{Z},$$

and therefore

1.1.2 Weak Perspective

$$\begin{cases} x = d\dfrac{X}{Z}, \\ y = d\dfrac{Y}{Z}. \end{cases} \quad (1.1)$$

As noted in the previous section, pinhole perspective is only an approximation of the geometry of the imaging process. This section discusses a coarser approximation, called weak perspective, which is also useful on occasion.

Consider the fronto-parallel plane Π_0 defined by $Z = Z_0$ (Figure 1.5). For any point P in Π_0 we can rewrite Eq. (1.1) as

$$\begin{cases} x = -mX, \\ y = -mY, \end{cases} \quad \text{where } m = -\frac{d}{Z_0}. \quad (1.2)$$

Physical constraints impose that Z_0 be negative (the plane must be in front of the pinhole), so the magnification m associated with the plane Π_0 is positive. This name is justified by the following remark: consider two points P and Q in Π_0

and their images p and q (Figure 1.5); obviously, the vectors $\overrightarrow{PQ}$ and $\overrightarrow{pq}$ are parallel, and we have $\|\overrightarrow{pq}\| = m\|\overrightarrow{PQ}\|$. This is the dependence of image size on object distance noted earlier.

j *d* *O*

q *P*

Π

k

Π_0

$-Z_0$

p *Q*

i

FIGURE 1.5: Weak-perspective projection. All line segments in the plane Π_0 are projected with the same magnification.

When a scene's relief is small relative to its average distance from the camera, the magnification can be taken to be constant. This projection model is called weak perspective, or scaled orthography.

When it is a priori known that the camera will always remain at a roughly constant distance from the scene, we can go further and normalize the image coor dinates so that m = −1. This is orthographic projection, defined by

$$x = X, \quad y = Y, \tag{1.3}$$

with all light rays parallel to the k axis and orthogonal to the image plane π (Figure 1.6). Although weak-perspective projection is an acceptable model for many imaging conditions, assuming pure orthographic projection is usually unrealistic.

j Π

p *P*

k

O

q *Q*

i

FIGURE 1.6: Orthographic projection. Unlike other geometric models of the image for

mation process, orthographic projection does not involve a reversal of image features. Accordingly, the magnification is taken to be negative, which is a bit unnatural but simplifies the projection equations.

1.1.3 Cameras with Lenses

Most real cameras are equipped with lenses. There are two main reasons for this: The first one is to gather light, because a single ray of light would otherwise reach each point in the image plane under ideal pinhole projection. Real pinholes have a finite size, of course, so each point in the image plane is illuminated by a cone of light rays subtending a finite solid angle. The larger the hole, the wider the cone and the brighter the image, but a large pinhole gives blurry pictures. Shrinking the pinhole produces sharper images but reduces the amount of light reaching the image plane, and may introduce diffraction effects. Keeping the picture in sharp focus while gathering light from a large area is the second main reason for using a lens.

Ignoring diffraction, interferences, and other physical optics phenomena, the behavior of lenses is dictated by the laws of geometric optics (Figure 1.7): (1) light travels in straight lines (light rays) in homogeneous media; (2) when a ray is reflected from a surface, this ray, its reflection, and the surface normal are coplanar, and the angles between the normal and the two rays are complementary; and (3) when a ray passes from one medium to another, it is refracted, i.e., its direction changes. According to Snell's law, if r_1 is the ray incident to the interface between two transparent materials with indices of refraction n_1 and n_2, and r_2 is the refracted ray, then r_1, r_2, and the normal to the interface are coplanar, and the angles α_1 and α_2 between the normal and the two rays are related by

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2. \quad (1.4)$$

α_1 $\quad {}_1r\ r'_1$

α_1 1

n

r 2

n_2

α_2

FIGURE 1.7: Reflection and refraction at the interface between two homogeneous media with indices of refraction n_1 and n_2.

In this chapter, we will only consider the effects of refraction and ignore those of reflection. In other words, we will concentrate on lenses as opposed to catadioptric optical systems (e.g., telescopes) that may include both reflective (mirrors) and refractive elements. Tracing light rays as they travel through a lens is simpler when the angles between these rays and the refracting surfaces of the lens are assumed to be small, which is the domain of paraxial (or first-order) geometric optics, and Snell's law becomes $n_1\alpha_1 \approx n_2\alpha_2$. Let us also assume that the lens is rotationally symmetric about a straight line, called its optical axis, and that all refractive surfaces are spherical. The symmetry of this setup allows us to determine

P

Y

F' F

O

f f -y

p

z -Z

FIGURE 1.8: A thin lens. Rays passing through O are not refracted. Rays parallel to the optical axis are focused on the focal point F .

the projection geometry by considering lenses with circular boundaries lying in a plane that contains the optical axis. In particular, consider a lens with two spherical surfaces of radius R and index of refraction n. We will assume that this lens is surrounded by vacuum (or, to an excellent approximation, by air), with an index of refraction equal to 1, and that it is thin, i.e., that a ray entering the lens and refracted at its right boundary is immediately refracted again at the left boundary.

Consider a point P located at (negative) depth Z off the optical axis, and denote by (PO) the ray passing through this point and the center O of the lens (Figure 1.8). It easily follows from the paraxial form of Snell's law that (PO) is not refracted, and that all the other rays passing through P are focused by the thin lens on the point p with depth z along (PO) such that

$$\frac{1}{z} - \frac{1}{Z} = \frac{1}{f}, \quad (1.5)$$

where $f = \frac{R}{2(n-1)}$ is the focal length of the lens.

Note that the equations relating the positions of P and p are exactly the same as under pinhole perspective projection if we take d = z since P and p lie on a ray passing through the center of the lens, but that points located at a distance −Z from O will be in sharp focus only when the image plane is located at a distance z from O on the other side of the lens that satisfies Eq. (1.5), the thin lens equation. Letting $Z \to -\infty$ shows that f is the distance between the center of the lens and the plane where objects such as stars (that are effectively located at $Z = -\infty$) focus. The two points F and F located at distance f from the lens center on the optical axis are called the focal points of the lens. In practice, objects within some range of distances (called depth of field or depth of focus) will be in acceptable focus. As shown in the problems at the end of this chapter, the depth of field increases with the f-number of the lens, i.e., the ratio between the focal length of the lens and its diameter.

Note that the field of view of a camera, i.e., the portion of scene space that actually projects onto the retina of the camera, is not defined by the focal length alone but also depends on the effective area of the retina (e.g., the area of film that can be exposed in a photographic camera, or the area of the sensor in a digital camera; see Figure 1.9).

film

a

φ lens

f

FIGURE 1.9: The field of view of a camera. It can be defined as 2φ, where $\varphi \stackrel{\text{def}}{=} \arctan \frac{a}{2f}$, a is the diameter of the sensor (film, CCD, or CMOS chip), and f is the focal length of the camera.

A more realistic model of simple optical systems is the thick lens. The equa tions describing its behavior are easily derived from the paraxial refraction equation, and they are the same as the pinhole perspective and thin lens projection equations, except for an offset (Figure 1.10). If H and H denote the principal points of the lens, then Eq. (1.5) holds when $-Z$ (resp. z) is the distance between P (resp. p) and the plane perpendicular to the optical axis and passing through H (resp. H). In this case, the only undeflected ray is along the optical axis.

P

Y

F' F H' H

-y

p

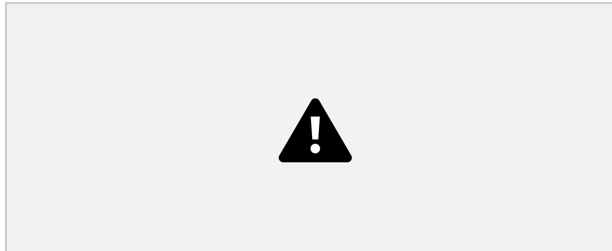
ff -Z

FIGURE 1.10: A simple thick lens with two spherical surfaces.

Simple lenses suffer from a number of aberrations. To understand why, let us remember that the paraxial refraction model is only an approximation, valid when the angle α between each ray along the optical path and the optical axis of the length is small and $\sin \alpha \approx \alpha$. This corresponds to a first-order Taylor expansion of the sine function. For larger angles, additional terms yield a better approximation, and it is easy to show that rays striking the interface farther from the optical axis are focused closer to the interface. The same phenomenon occurs for a lens, and it is the source of two types of spherical aberrations (Figure 1.11 [a]): Consider a point P on the optical axis and its paraxial image p. The distance between p and the intersection of the optical axis with a ray issued from P and refracted by the lens is called the longitudinal spherical aberration of that ray. Note that if an image plane Π were erected in P, the ray would intersect this plane at some distance from

the axis, called the transverse spherical aberration of that ray. Together, all rays passing through P and refracted by the lens form a circle of confusion centered in P as they intersect Π. The size of that circle will change if we move Π along the optical axis. The circle with minimum diameter is called the circle of least confusion, and its center does not coincide (in general) with p.

e
p P



(a) (b)

(c)
Π

FIGURE 1.11: Aberrations. (a) Spherical aberration: The gray region is the paraxial zone where the rays issued from P intersect at its paraxial image p. If an image plane π were erected in p, the image of p in that plane would form a circle of confusion of diameter e. The focus plane yielding the circle of least confusion is indicated by a dashed line. (b) Distortion: From left to right, the nominal image of a fronto-parallel square, pincushion distortion, and barrel distortion. (c) Chromatic aberration: The index of refraction of a transparent medium depends on the wavelength (or color) of the incident light rays. Here, a prism decomposes white light into a palette of colors. Figure from US NAVY MANUAL OF BASIC OPTICS AND OPTICAL INSTRUMENTS, prepared by the Bureau of Naval Personnel, reprinted by Dover Publications, Inc. (1969).

Besides spherical aberration, there are four other types of primary aberrations caused by the differences between first- and third-order optics, namely coma, astig matism, field curvature, and distortion. A precise definition of these aberrations is beyond the scope of this book. Suffice to say that, like a spherical aberration, the first three degrade the image by blurring the picture of every object point. Distor tion, on the other hand, plays a different role and changes the shape of the image

FIGURE 1.12: Vignetting effect in a two-lens system. The shaded part of the beam never reaches the second lens. Additional apertures and stops in a lens further contribute to vignetting.

as a whole (Figure 1.11 [b]). This effect is due to the fact that different areas of a lens have slightly different focal lengths. The aberrations mentioned so far are monochromatic, i.e., they are independent of the response of the lens to various wavelengths. However, the index of refraction of a transparent medium depends on wavelength (Figure 1.11 [c]), and it follows from the thin lens equation (Eq. [1.5]) that the focal length depends on wavelength as well. This causes the phenomenon of chromatic aberration: refracted rays corresponding to different wavelengths will intersect the optical axis at different points (longitudinal chromatic aberration) and form different circles of confusion in the same image plane (transverse chromatic aberration).

Aberrations can be minimized by aligning several simple lenses with well chosen shapes and refraction indices, separated by appropriate stops. These com pound lenses can still be modeled by the thick lens equations, but they suffer from one more defect relevant to machine vision: light beams emanating from object points located off-axis are partially blocked by the various apertures (including the individual lens components themselves) positioned inside the lens to limit aberra tions (Figure 1.12). This phenomenon, called vignetting, causes the image bright ness to drop in the image periphery. Vignetting may pose problems to automated image analysis programs, but it is not quite as important in photography, thanks to the human eye’s remarkable insensitivity to smooth brightness gradients. Speaking of which, it is time to have a look at this extraordinary organ.

1.1.4 The Human Eye

Here we give a (brief) overview of the anatomical structure of the eye. It is largely based on the presentation in Wandell (1995), and the interested reader is invited to read this excellent book for more details. Figure 1.13 (left) is a sketch of the section of an eyeball through its vertical plane of symmetry, showing the main elements of the eye: the iris and the pupil, which control the amount of light penetrating the eyeball; the cornea and the crystalline lens, which together refract the light to create the retinal image; and finally the retina, where the image is

formed. Despite its globular shape, the human eyeball is functionally similar to a camera with a field of view covering a 160° (width) × 135° (height) area. Like any other optical system, it suffers from various types of geometric and chromatic aberrations. Several models of the eye obeying the laws of first-order geometric optics have been proposed, and Figure 1.13 (right) shows one of them, Helmoltz’s schematic eye. There are only three refractive surfaces, with an infinitely thin cornea and a homogeneous lens. The constants given in Figure 1.13 are for the eye focusing at infinity (unaccommodated eye). This model is of course only an approximation of the real optical characteristics of the eye.

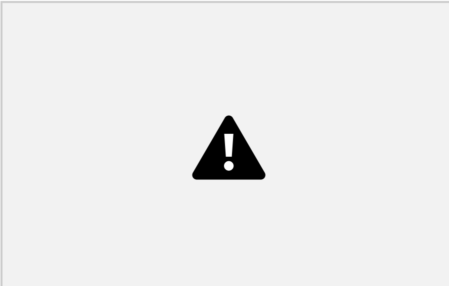


FIGURE 1.13: Left: the main components of the human eye. Reproduced with permission, the American Society for Photogrammetry and Remote Sensing. A.L. Nowicki, “Stere oscopy.” MANUAL OF PHOTOGRAMMETRY, edited by M.M. Thompson, R.C. Eller, W.A. Radlinski, and J.L. Speert, third edition, pp. 515–536. Bethesda: American Society of Photogrammetry, (1966). Right: Helmoltz’s schematic eye as modified by Laurance (after Driscoll and Vaughan, 1978). The distance between the pole of the cornea and the anterior principal plane is 1.96 mm, and the radii of the cornea, anterior, and posterior surfaces of the lens are respectively 8 mm, 10 mm, and 6 mm.

Let us have a second look at the components of the eye one layer at a time. The cornea is a transparent, highly curved, refractive window through which light enters the eye before being partially blocked by the colored and opaque surface of the iris. The pupil is an opening at the center of the iris whose diameter varies from about 1 to 8 mm in response to illumination changes, dilating in low light to increase the amount of energy that reaches the retina and contracting in normal lighting conditions to limit the amount of image blurring due to spherical aberration in the eye. The refracting power (reciprocal of the focal length) of the eye is, in large part, an effect of refraction at the the air–cornea interface, and it is fine-tuned by deformations of the crystalline lens that accommodates to bring objects into sharp focus. In healthy adults, it varies between 60 (unaccommodated case) and 68 diopters (1 diopter = 1 m^{-1}), corresponding to a range of focal lengths between 15 and 17 mm.

The retina itself is a thin, layered membrane populated by two types of photoreceptors—rods and cones. There are about 100 million rods and 5 million cones in a human eye. Their spatial distribution varies across the retina: The mac ula lutea is a region in the center of the retina where the concentration of cones is particularly high and images are sharply focused whenever the eye fixes its attention on an object (Figure 1.13). The highest concentration of cones occurs in the fovea,

a depression in the middle of the macula lutea where it peaks at $1.6 \times 10^5/mm^2$, with the centers of two neighboring cones separated by only half a minute of visual angle. Conversely, there are no rods in the center of the fovea, but the rod density increases toward the periphery of the visual field. There is also a blind spot on the retina, where the ganglion cell axons exit the retina and form the optic nerve.

The rods are extremely sensitive photoreceptors, capable of responding to a single photon, but they yield relatively poor spatial detail despite their high number because many rods converge to the same neuron within the retina. In contrast, cones become active at higher light levels, but the signal output by each cone in the fovea is encoded by several neurons, yielding a high resolution in that area. As discussed further in Chapter 3, there are three types of cones with different spectral sensitivities, and these play a key role in the perception of color. Much more could (and should) be said about the human eye—for example how our two eyes verge and fixate on targets, and how they cooperate in stereo vision, an issue briefly discussed in Chapter 7.

1.2 INTRINSIC AND EXTRINSIC PARAMETERS

Digital images, like animal retinas, are spatially discrete, and divided into (usually) rectangular picture elements, or pixels. This is an aspect of the image formation process that we have neglected so far, assuming instead that the image domain is spatially continuous. Likewise, the perspective equation derived in the previous section is valid only when all distances are measured in the camera's reference frame, and when image coordinates have their origin at the image center where the axis of symmetry of the camera pierces its retina. In practice, the world and camera coordinate systems are related by a set of physical parameters, such as the focal length of the lens, the size of the pixels, the position of the image center, and the position and orientation of the camera. This section identifies these parameters. We will distinguish the intrinsic parameters, which relate the camera's coordinate system to the idealized coordinate system used in Section 1.1, from the extrinsic parameters, which relate the camera's coordinate system to a fixed world coordinate system and specify its position and orientation in space.

We ignore in the rest of this section the fact that, for cameras equipped with a lens, a point will be in focus only when its depth and the distance between the optical center of the camera and its image plane obey Eq. (1.5). In particular, we assume that the camera is focused at infinity, so d = f. Likewise, the nonlinear aberrations associated with real lenses are not taken into account by Eq. (1.1). We neglect these aberrations in this section, but revisit radial distortion in Section 1.3 when we address the problem of estimating the intrinsic and extrinsic parameters of a camera (a process known as geometric camera calibration).

1.2.1 Rigid Transformations and Homogeneous Coordinates

This section features our first use of homogeneous coordinates to represent the position of points in two or three dimensions. Consider a point P whose position in some coordinate frame (F)=(O, i, j, k) is given by

$$\overrightarrow{OP} = Xi + Yj + Zk.$$

We define the usual (nonhomogeneous) coordinate vector of P to be the vector $(X, Y, Z)^T$ in R^3 and its homogeneous coordinate vector as the vector $(X, Y, Z, 1)^T$ in R^4. We use bold letters to denote (homogeneous and nonhomogeneous) coordi nate vectors in this book, and always state which type of coordinates we use when it is not obvious from the context. We also use a superscript on the left side of coordinate vectors when necessary to indicate which coordinate frame a position is expressed in. For example, ${}^FP$ stands for the coordinate vector of the point P in the frame (F). Homogeneous coordinates are a convenient device for representing various geometric transformations by matrix products. For example, the change of coordinates between two Euclidean coordinate systems (A) and (B) may be rep resented by a 3 × 3 rotation matrix R and a translation vector t in R^3, and the corresponding rigid transformation can be written in nonhomogeneous coordinates as

$${}^AP = R\,{}^BP + t, \quad (1.6)$$

where ${}^AP$ and ${}^BP$ are elements of R^3. In homogeneous coordinates, we write instead

${}^AP = T\,{}^BP$, where $T =$ time elements of R^4. $R\ t\ 0^T\ 1$, (1.7)

and ${}^AP$ and ${}^BP$ are this

Before going further, let us recall a few facts about rotations. Rotation matri ces form a mulitplicative group. From an analytical viewpoint, they are character ized by the facts that (1) the inverse of a rotation matrix is equal to its transpose, and (2) its determinant is equal to one. It can also be shown that any rotation matrix can be parameterized by three Euler angles, or written as the product of three elementary rotations about the i, j, and k vectors of some coordinate sys tem. As shown in Chapters 7 and 14, other parameterizations—by exponentials of antisymmetric matrices or quaternions for example—may prove useful as well. Ge ometrically, the matrix R in Eq. (1.6) also represents the basis vectors (i_B, j_B, k_B) of (B) in the coordinate frame (A)—that is, the matrix R in Eq. (1.6) is given by:

$$= {}^Ai_B,\ {}^Aj_B,\ {}^Ak_B = R \stackrel{\text{def}}{} \quad \begin{matrix} i_A \cdot i_B\ j_A \cdot i_B\ k_A \cdot i_B \\ i_A \cdot j_B\ j_A \cdot j_B\ k_A \cdot j_B\ i_A \cdot k_B \\ j_A \cdot k_B\ k_A \cdot k_B \end{matrix}, \quad (1.8)$$

and, as shown in the problems at the end of this chapter, Eq. (1.6) easily follows from this definition. By definition, the columns of a rotation matrix form a right handed orthonormal coordinate system of R^3. It follows from properties (1) and (2) that their rows also form such a coordinate system. One may wonder what happens when R is replaced in Eq. (1.7) by some arbitrary nonsingular 3×3 matrix, or when the matrix T itself is replaced by some arbitrary nonsingular 4×4 matrix. As further discussed in Chapter 8, the coordinate frames (A) and (B) are no longer separated by rigid transformations in this case, but by affine and projective transformations respectively.

As will be shown in the rest of this section, homogeneous coordinates also provide an algrebraic representation of the perspective projection process in the form of a 3×4 matrix M, so that the coordinate vector $P = (X, Y, Z, 1)^T$ of a point P in some fixed world coordinate system and the coordinate vector $p = (x, y, 1)^T$ of

j

y

P p

p

y

k

$_0$ *c*

Pinhole

c *x*

i

Normalized
image plane

Physical
retina

c

x

FIGURE 1.14: Physical and normalized image coordinate systems.

its image p in the camera's reference frame are related by the perspective projection equation

$$p = \frac{1}{Z}\mathcal{M}P. \quad (1.9)$$

1.2.2 Intrinsic Parameters

It is possible to associate with a camera a normalized image plane parallel to its physical retina but located at a unit distance from the pinhole. We attach to this plane its own coordinate system with an origin located at the point $\hat{c}$ where the optical axis pierces it (Figure 1.14). Equation (1.1) can be written in this normalized coordinate system as

$$\begin{cases} \hat{x} = \dfrac{X}{Z} \\ \hat{y} = \dfrac{Y}{Z} \end{cases} \iff \hat{p} = \frac{1}{Z}\begin{pmatrix}\mathrm{Id} & 0\end{pmatrix} P, \quad (1.10)$$

where $\hat{p} \stackrel{\text{def}}{=} (\hat{x}, \hat{y}, 1)^T$ is the vector of homogeneous coordinates of the projection $\hat{p}$ of the point P into the normalized image plane, and P is as before the homogeneous coordinate vector of P in the world coordinate frame.

The physical retina of the camera is in general different (Figure 1.14): It is located at a distance $f = 1$ from the pinhole (remember that we assume that the camera is focused at infinity, so the distance between the pinhole and the image plane is equal to the focal length), and the coordinates (x, y) of the image point p are usually expressed in pixel units (instead of, say, meters). In addition, pixels may be rectangular instead of square, so the camera has two additional scale parameters k and l, and

$$\begin{cases} x = kf\dfrac{X}{Z} = kf\hat{x}, \\ y = lf\dfrac{Y}{Z} = lf\hat{y}. \end{cases} \quad (1.11)$$

Let us talk units for a second: f is a distance, expressed in meters, for example, and a pixel will have dimensions $\frac{1}{k} \times \frac{1}{l}$, where k and l are expressed in pixel$\times m^{-1}$. The parameters k, l, and f are not independent, and they can be replaced by the magnifications $\alpha = kf$ and $\beta = lf$ expressed in pixel units.

Now, in general, the actual origin of the camera coordinate system is at a corner c of the retina (in the case depicted in Figure 1.14, the lower-left corner, or sometimes the upper-left corner, when the image coordinates are the row and column indices of a pixel) and not at its center, and the center of the CCD matrix usually does not coincide with the image center c_0. This adds two parameters x_0 and y_0 that define the position (in pixel units) of c_0 in the retinal coordinate system. Thus, Eq. (1.11) is replaced by

$$\begin{cases} x = \alpha\hat{x} + x_0, \\ y = \beta\hat{y} + y_0. \end{cases} \quad (1.12)$$

Finally, the camera coordinate system might also be skewed, due to some manufacturing error, so the angle θ between the two image axes is not equal to (but of course not very different from) 90 degrees. In this case, it is easy to show that Eq. (1.12) transforms into

$$\begin{cases} x = \alpha\hat{x} - \alpha\cot\theta\hat{y} + x_0, \\ y = \dfrac{\beta}{\sin\theta}\hat{y} + y_0. \end{cases} \quad (1.13)$$

This can be written in matrix form as

$$p = K\hat{p}, \quad \text{where } p = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \text{ and } K \stackrel{\text{def}}{=} \begin{pmatrix} \alpha & -\alpha\cot\theta & x_0 \\ 0 & \dfrac{\beta}{\sin\theta} & y_0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (1.14)$$

The 3×3 matrix K is called the (internal) calibration matrix of the camera. Putting Eqs. (1.10) and (1.14) together, we obtain

$$p = \frac{1}{Z}K\begin{pmatrix} \mathrm{Id} & \mathbf{0} \end{pmatrix} P = \frac{1}{Z}\mathcal{M}P, \quad \text{where } \mathcal{M} \stackrel{\text{def}}{=} \begin{pmatrix} K & \mathbf{0} \end{pmatrix}, \quad (1.15)$$

which is indeed an instance of Eq. (1.9). The five parameters α, β, θ, x_0, and y_0 are called the intrinsic parameters of the camera.

Several of these parameters, such as the focal length, or the physical size of the pixels, are often available in the EXIF tags attached to the JPEG images recorded by digital cameras (this information might not be available, of course, as in the case of stock film footage). For zoom lenses, the focal length may vary with

time, along with the image center when the optical axis of the lens is not exactly perpendicular to the image plane. Simply changing the focus of the camera will also affect the magnification because it will change the lens-to-retina distance, but we will continue to assume that the camera is focused at infinity and ignore this effect in the rest of this chapter.

1.2.3 Extrinsic Parameters

Equation (1.15) is written in a coordinate frame (C) attached to the camera. Let us now consider the case where this frame is distinct from the world coordinate system (W). To emphasize this, we rewrite Eq. (1.15) as $p = \frac{1}{Z}\mathcal{M}\,{}^{C}P$, where ${}^{C}P$ denotes the vector of homogeneous coordinates of the point P expressed in (C). The change of coordinates between (C) and (W) is a rigid transformation, and it can be written as

$${}^{C}P = \begin{pmatrix} \mathcal{R} & t \\ \mathbf{0}^T & 1 \end{pmatrix} {}^{W}P,$$

where ${}^{W}P$ is the vector of homogeneous coordinates of the point P in the coordinate frame (W). Taking $P = {}^{W}P$ and substituting in Eq. (1.15) finally yields

$$p = \frac{1}{Z}\mathcal{M}P, \quad \text{where } \mathcal{M} = K\begin{pmatrix} \mathcal{R} & t \end{pmatrix}. \quad (1.16)$$

This is the most general form of the perspective projection equation, and indeed an instance of Eq. (1.9). Knowing M determines the position of the camera's optical center in the coordinate frame (W)—that is, its homogeneous coordinate vector $O = {}^{W}O$. Indeed, as shown in the problems at the end of this chapter, $MO = 0$.

As mentioned earlier, a rotation matrix such as R is defined by three indepen dent parameters (for example, Euler angles). Adding to these the three coordinates of the vector t, we obtain a set of six extrinsic parameters that define the position and orientation of the camera relative to the world coordinate frame.

It is very important to understand that the depth Z in Eq. (1.16) is not independent of M and P , because if m^T_1 , m^T_2 and m^T_3 denote the three rows of M, it follows directly from Eq. (1.16) that $Z = m_3 \cdot P$. In fact, it is sometimes convenient to rewrite Eq. (1.16) in the equivalent form:

$$x = \frac{m_1 \cdot P}{m_3 \cdot P}, \quad y = \frac{m_2 \cdot P}{m_3 \cdot P}. \quad (1.17)$$

A perspective projection matrix can be written explicitly as a function of its five intrinsic parameters, the three rows r^T_1 , r^T_2 , and r^T_3 of the matrix R, and the three coordinates t_1, t_2, and t_3 of the vector t, namely:

$$M = \begin{pmatrix} \alpha r^T_1 - \alpha \cot\theta\, r^T_2 + x_0 r^T_3 & \alpha t_1 - \alpha \cot\theta\, t_2 + x_0 t_3 \\ \frac{\beta}{\sin\theta} r^T_2 + y_0 r^T_3 & \frac{\beta}{\sin\theta} t_2 + y_0 t_3 \\ r^T_3 & t_3 \end{pmatrix}. \quad (1.18)$$

When R is written as the product of three elementary rotations, the vectors r_i ($i = 1, 2, 3$) can of course be written in terms of the corresponding three angles, and Eq. (1.18) gives an explicit parameterization of M in terms of all 11 camera parameters.

1.2.4 Perspective Projection Matrices

This section examines the conditions under which a 3×4 matrix M can be written in the form given by Eq. (1.18). Let us write without loss of generality $M = (A \; b)$, where A is a 3 × 3 matrix and b is an element of $\mathbb{R}^3$, and let us denote by a^T_3 the third row of A. Clearly, if M is an instance of Eq. (1.18), then a^T_3 must be a unit vector since it is equal to r^T_3 , the last row of a rotation matrix. Note, however, that replacing M by λM in Eq. (1.17) for some arbitrary $\lambda = 0$ does not change the corresponding image coordinates. This will lead us in the rest of this book to consider projection matrices as homogeneous objects, only defined up to scale, whose canonical form, as expressed by Eq. (1.18), can be obtained by choosing a scale factor such that $||a_3|| = 1$. Note that the parameter Z in Eq. (1.16) can only rightly be interpreted as the depth of the point P when M is written in this canonical form. Note also that the number of intrinsic and extrinsic parameters of a camera matches the 11 free parameters of the (homogeneous) matrix M.

We say that a 3×4 matrix that can be written (up to scale) as Eq. (1.18) for some set of intrinsic and extrinsic parameters is a perspective projection matrix. It is of practical interest to put some restrictions on the intrinsic parameters of a camera because, as noted earlier, some of these parameters will be fixed and might be known. In particular, we will say that a 3×4 matrix is a zero-skew perspective projection matrix when it can be rewritten (up to scale) as Eq. (1.18) with $\theta = \pi/2$, and that it is a perspective projection matrix with zero skew and unit aspect-ratio when it can be rewritten (up to scale) as Eq. (1.18) with $\theta = \pi/2$ and $\alpha = \beta$.A camera with known nonzero skew and nonunit aspect-ratio can be transformed into a camera with zero skew and unit aspect-ratio by an appropriate change of image coordinates. Are arbitrary 3×4 matrices perspective projection matrices? The following theorem answers this question.

Theorem 1. Let $M = A\ b$ be a 3×4 matrix, and let a^T_i (i =1, 2, 3) denote the rows of the matrix A formed by the three leftmost columns of M.

- A necessary and sufficient condition for M to be a perspective projection matrix is that Det(A) = 0.
- A necessary and sufficient condition for M to be a zero-skew perspective projection matrix is that Det(A) = 0 and

$$(a_1 \times a_3) \cdot (a_2 \times a_3) = 0.$$

- A necessary and sufficient condition for M to be a perspective projection matrix with zero skew and unit aspect-ratio is that Det(A) = 0 and

$$(a_1 \times a_3) \cdot (a_2 \times a_3) = 0,$$
$$(a_1 \times a_3) \cdot (a_1 \times a_3) = (a_2 \times a_3) \cdot (a_2 \times a_3).$$

Q
O

p q r

P P'

R

Q'

Π *r*

R

p

Q'

q

Q

r

O

Π

P P'

r

FIGURE 1.15: Affine projection models: (top) weak-perspective and (bottom) paraper spective projections.

The conditions of the theorem are clearly necessary: By definition, given some perspective projection matrix A, we can always write ρA = KR for some nonzero scalar ρ, calibration matrix K, rotation matrix R, and vector t. In particular, ρ^3Det(A) = Det(K) = 0 since calibration matrices are nonsingular by construction, so A is nonsingular. Further, a simple calculation shows that the rows of the matrix 1

$_\rho$KR satisfy the conditions of the theorem under the various assumptions imposed by its statement. These conditions are proven to also be sufficient in Faugeras (1993).

1.2.5 Weak-Perspective Projection Matrices

As noted in Section 1.1.2, when a scene's relief is small compared to the overall distance separating it from the camera observing it, a weak-perspective projection model can be used to approximate the imaging process (Figure 1.15, top). Let O denote the optical center of the camera, and let R denote a scene reference point. The weak-perspective projection of a scene point P is constructed in two steps: the point P is first projected orthogonally onto a point P of the plane Π_r parallel to the image plane Π and passing through R; perspective projection is then used to map the point P onto the image point p. Since π_r is a fronto-parallel plane, the net effect of the second projection step is a scaling of the image coordinates.

As shown in this section, the weak-perspective projection process can be rep resented in terms of a 2 × 4 matrix M, so that the homogeneous coordinate vector P =(X, Y, Z, 1)^T of a point P in some fixed world coordinate system and the non homogeneous coordinate vector p =(x, y)^T of its image p in the camera's reference

frame are related by the affine projection equation

$$p = MP . \quad (1.19)$$

It turns out that this general model accomodates various other approximations of the perspective projection process. These include the orthographic projection model discussed earlier, as well as the parallel projection model, which subsumes the orthographic one, and takes into account the fact that the objects of interest may lie off the optical axis of the camera. In this model, the viewing rays are parallel to each other but not necessarily perpendicular to the image plane. Paraperspective is another affine projection model that takes into account both the distortions associated with a reference point that is off the optical axis of the camera and possible variations in depth (Figure 1.15, bottom). Using the same notation as before, and denoting by Δ the line joining the optical center O to the reference point R, parallel projection in the direction of Δ is first used to map P onto a point P of the plane Π_r; perspective projection is then used to map the point P onto the image point p.

We will focus on weak perspective in the rest of this section. Let us derive the corresponding projection equation. If Z_r denotes the depth of the reference point R, the two elementary projection stages P → P → p can be written in the normalized coordinate system attached to the camera as

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \longrightarrow \begin{pmatrix} X \\ Y \\ Z_r \end{pmatrix} \longrightarrow \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = \begin{pmatrix} X/Z_r \\ Y/Z_r \\ 1 \end{pmatrix},$$

or, in matrix form,

$$\begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = \frac{1}{Z_r} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & Z_r \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}.$$

Introducing the calibration matrix K of the camera and its extrinsic parameters R and t gives the general form of the projection equation, i.e.,

$$p = \frac{1}{Z_r} K \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & Z_r \end{pmatrix} \begin{pmatrix} R & t \\ 0^T & 1 \end{pmatrix} P, \quad (1.20)$$

where P and p denote as before the homogeneous coordinate vector of the point P in the world reference frame, and the homogeneous coordinate vector of its projection p in the camera's coordinate system. Finally, noting that Z_r is a constant and writing

$$K = \begin{pmatrix} K_2 & p_0 \\ 0^T & 1 \end{pmatrix}, \text{ where } K_2 \stackrel{def}{=} \begin{pmatrix} \alpha & -\alpha \cot\theta \\ 0 & \frac{\beta}{\sin\theta} \end{pmatrix} \text{ and } p_0 \stackrel{def}{=} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix},$$

allows us to rewrite Eq. (1.20) as

$$p = \mathcal{M}P, \text{ where } \mathcal{M} = \begin{pmatrix} A & b \end{pmatrix}, \quad (1.21)$$

where p is, this time, the nonhomogeneous coordinate vector of the point p, and $\mathcal{M}$ is a 2 × 4 projection matrix (compare to the general perspective case of Eq. [1.16]). In this expression, the 2 × 3 matrix A and the 2-vector b are respectively defined by

$$A = \frac{1}{Z_r} K_2 R_2 \text{ and } b = \frac{1}{Z_r} K_2 t_2 + p_0,$$

where R_2 denotes the 2×3 matrix formed by the first two rows of R, and t_2 denotes the 2-vector formed by the first two coordinates of t.

Note that t_3 does not appear in the expression of $\mathcal{M}$, and that t_2 and p_0 are coupled in this expression: the projection matrix does not change when t_2 is replaced by $t_2 + a$ and p_0 is replaced by $p_0 - \frac{1}{Z_r} K_2 a$. This redundancy allows us to arbitrarily choose $x_0 = y_0 = 0$. In other words, the position of the center of the image is immaterial for weak-perspective projection. Note that the values of Z_r, α, and β are also coupled in the expression of $\mathcal{M}$, and that the value of Z_r is a priori unknown in most applications. This allows us to write

$$\mathcal{M} = \frac{1}{Z_r} \begin{pmatrix} k & s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_2 & t_2 \end{pmatrix}, \quad (1.22)$$

where k and s denote the aspect ratio and the skew of the camera, respectively. In particular, a weak-perspective projection matrix is defined by two intrinsic param eters (k and s), five extrinsic parameters (the three angles defining R_2 and the two coordinates of t_2), and one scene-dependent structure parameter Z_r.

A 2×4 matrix $M = \begin{pmatrix} A & b \end{pmatrix}$ where A is an arbitrary rank-2 2×3 matrix and b is an arbitrary vector in $\mathbb{R}^2$ is called an affine projection matrix. Both weak perspective and general affine projection matrices are defined by eight independent parameters. Weak-perspective projection matrices are affine ones of course. Con versely, a simple parameter-counting argument suggests that it should be possible to write an arbitrary affine projection matrix as a weak-perspective one. This is confirmed by the following theorem.

Theorem 2. An affine projection matrix can be written uniquely (up to a sign ambiguity) as a general weak-perspective projection matrix as defined by Eq. (1.22).

This theorem is proven in Faugeras et al. (2001, Propositions 4.26 and 4.27) and the problems.

1.3 GEOMETRIC CAMERA CALIBRATION

This section addresses the problem of estimating the intrinsic and extrinsic param eters of a camera from the image positions of scene features such as points of lines, whose positions are known in some fixed world coordinate system (Figure 1.16). In this context, camera calibration can be modeled as an optimization process, where the discrepancy between the observed image features and their theoretical positions is minimized with respect to the camera's intrinsic and extrinsic parameters.

Specifically, we assume that the image positions (x_i, y_i) of n fiducial points P_i $(i = 1, \ldots, n)$ with known homogeneous coordinate vectors P_i have been found

(W)

P_i

O_W

k_W k_C p_i

(C)

j_C

j_W

i_C i_W

FIGURE 1.16: Camera calibration setup: In this example, the calibration rig is formed by three grids drawn in orthogonal planes. Other patterns could be used as well, and they may involve lines or other geometric figures.

in a picture of a calibration rig, either automatically or by hand. In the absence of modeling and measurement errors, geometric camera calibration amounts to finding the intrinsic and extrinsic parameters ξ such that

$$\begin{cases} x_i = \dfrac{m_1(\xi) \cdot P_i}{m_3(\xi) \cdot P_i}, \\[2ex] y_i = \dfrac{m_2(\xi) \cdot P_i}{m_3(\xi) \cdot P_i}, \end{cases} \quad (1.23)$$

where $m^T_i(\xi)$ denotes the ith row of the projection matrix M, explicitly parameterized in this equation by the camera parameters. In the typical case where there are more measurements than unknowns (at least six points for 11 intrinsic and extrin sic parameters), Eq. (1.23) does not admit an exact solution, and an approximate one has to be found as the solution of a least-squares minimization problem (see Chapter 22). We present two least-squares formulations of the calibration problem in the rest of this section. The corresponding algorithms are illustrated with the calibration data shown in Figure 1.17.

1.3.1 A Linear Approach to Camera Calibration

We decompose the calibration process into (1) the computation of the perspective projection matrix M associated with the camera, followed by (2) the estimation of the intrinsic and extrinsic parameters of the camera from this matrix.

Estimation of the Projection Matrix. Let us assume that our camera has nonzero skew. According to Theorem 1, the matrix M is not singular, but other wise arbitrary. Clearing the denominators in Eq. (1.23) yields two linear equations in m_1, m_2, and m_3 (we omit the parameters ξ from now on for the sake of con-

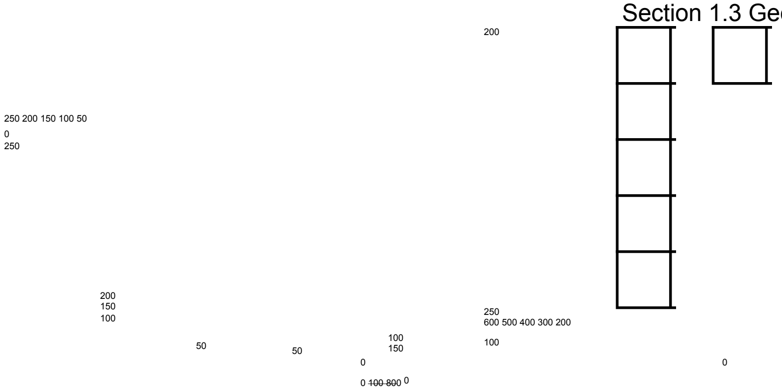


FIGURE 1.17: Camera calibration data. Left: A rendering of 491 3D fiducial points measured on a calibration rig. Right: The corresponding image points. Data courtesy of Janne Heikkil¨a; data copyright c 2000 University of Oulu.

ciseness), namely

$$(m_1 - x_i m_3) \cdot P_i = P^T_i m_1 + 0^T m_2 - x_i P^T_i m_3 = 0, \quad (m_2 - y_i m_3) \cdot P_i = 0^T m_1 + P^T_i m_2 - y_i P^T_i m_3 = 0.$$

Collecting the constraints associated with all points yields a system of 2n homogeneous linear equations in the 12 coefficients of the matrix M, namely,

$$Pm = 0, \quad (1.24)$$

where

$$P \stackrel{def}{=} \begin{pmatrix} P^T_1 & 0^T & -x_1 P^T_1 \\ 0^T & P^T_1 & -y_1 P^T_1 \\ \dots & \dots & \dots \\ P^T_n & 0^T & -x_n P^T_n \\ 0^T & P^T_n & -y_n P^T_n \end{pmatrix} \quad \text{and} \quad m \stackrel{def}{=} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} = 0.$$

When $n \geq 6$, homogeneous linear least-squares can be used to compute the value of the unit vector m (hence the matrix M) that minimizes $||Pm||^2$ as the eigenvector of the 12×12 matrix P^TP associated with its smallest eigenvalue (see Chapter 22). Note that any nonzero multiple of the vector m would have done just as well, reflecting the fact that M is defined by only 11 independent parameters.

Degenerate Point Configurations. Before showing how to recover the intrinsic and extrinsic parameters of the camera, let us pause to examine the degenerate configurations of the points P_i ($i = 1,...,n$) that may cause the failure of the camera calibration process. We focus on the (ideal) case where the positions p_i ($i = 1,...,n$) of the image points can be measured with zero error, and identify the nullspace of the matrix P (i.e., the subspace of R^{12} formed by the vectors l such that $Pl = 0$).

Let l be such a vector. Introducing the vectors formed by successive quadru ples of its coordinates—that is, $\lambda = (l_1,l_2,l_3,l_4)^T$, $\mu = (l_5,l_6,l_7,l_8)^T$, and

$v = (l_9,l_{10},l_{11},l_{12})^T$—allows us to write

$$0 = Pl = \begin{pmatrix} P_1^T & 0^T & -x_1P_1^T \\ 0^T & P_1^T & -y_1P_1^T \\ \dots & \dots & \dots \\ P_n^T & 0^T & -x_nP_n^T \\ 0^T & P_n^T & -y_nP_n^T \end{pmatrix} \begin{pmatrix} \lambda \\ \mu \\ v \end{pmatrix} = \begin{pmatrix} P_1^T\lambda - x_1P_1^Tv \\ P_1^T\mu - y_1P_1^Tv \\ \dots \\ P_n^T\lambda - x_nP_n^Tv \\ P_n^T\mu - y_nP_n^Tv \end{pmatrix}. \quad (1.25)$$

Combining Eq. (1.23) with Eq. (1.25) yields

$$\begin{cases} P_i^T\lambda - \dfrac{m_1^TP_i}{m_3^TP_i}P_i^Tv = 0, \\ P_i^T\mu - \dfrac{m_2^TP_i}{m_3^TP_i}P_i^Tv = 0, \end{cases} \quad \text{for } i = 1, \dots, n.$$

Thus, after clearing the denominators and rearranging the terms, we finally obtain:

$$\begin{cases} P_i^T(\lambda m_3^T - m_1v^T)P_i = 0, \\ P_i^T(\mu m_3^T - m_2v^T)P_i = 0, \end{cases} \quad \text{for } i = 1, \dots, n. \quad (1.26)$$

As expected, the vector l associated with $\lambda = m_1$, $\mu = m_2$, and $v = m_3$ is a solution of these equations. Are there other solutions?

Let us first consider the case where the points P_i ($i = 1,...,n$) all lie in some plane Π, so $P_i \cdot \Pi = 0$ for some 4-vector Π. Clearly, choosing (λ, μ, v) equal to $(\Pi, 0, 0)$, $(0, \Pi, 0)$, $(0, 0, \Pi)$, or any linear combination of these vectors will yield a solution of Eq. (1.26). In other words, the nullspace of P contains the four dimensional vector space spanned by these vectors and m. In practice, this means that the fiducial points P_i should not all lie in the same plane.

In general, for a given nonzero value of the vector l, the points P_i that satisfy Eq. (1.26) must lie on the curve where the two quadric surfaces defined by the corresponding equations intersect. A closer look at Eq. (1.26) reveals that the straight line where the planes defined by $m_3 \cdot P = 0$ and $v \cdot P = 0$ intersect lies on both quadrics. It can be shown that the intersection curve of these two surfaces consists of this line and of a twisted cubic curve Γ passing through the origin. A twisted cubic is determined entirely by six points lying on it, and it follows that seven points chosen at random will not fall on Γ. In addition, since this curve passes through the origin, choosing $n \geq 6$ random points will in general guarantee that the matrix P has rank 11 and that the projection matrix can be recovered in a unique fashion.

Estimation of the Intrinsic and Extrinsic Parameters. Once the projection matrix M has been estimated, its expression in terms of the camera's intrinsic and extrinsic parameters (Eq. [1.18]) can be used to recover these parameters as follows:

We write as before $M = (A \; b)$, with a^T_1, a^T_2, and a^T_3 denoting the rows of A, and obtain

a^T_2 r^T_3

a^T_3

$\rho (A \; b) = K(R \; t) \iff \rho$ $(a^T_1$ $) =$ $\alpha r^T_1 - \alpha \cot \theta r^T_2 + x_0 r^T_3$ $\frac{\beta}{\sin \theta} r^T_2 + y_0 r^T_3$ $),$

where ρ is an unknown scale factor, introduced here to account for the fact that the recovered matrix M has unit Frobenius form since $||M||_F = ||m|| = 1$. In particular, using the fact that the rows of a rotation matrix have unit length and are perpendicular to each other yields immediately

where $\varepsilon = \mp 1$.

$\rho = \varepsilon / ||a_3||,$

$r_3 = \rho a_3,$

(1.27) $x_0 = \rho^2 (a_1 \cdot a_3), \; y_0 = \rho^2 (a_2 \cdot a_3),$

Since θ is always in the neighborhood of π/2 with a positive sine, we have

$\rho^2 (a_1 \times a_3) = -\alpha r_2 - \alpha \cot \theta r_1,$

$\rho^2 (a_2 \times a_3) = \frac{\beta}{\sin \theta} r_1,$ and

$\rho^2 ||a_1 \times a_3|| = \frac{|\alpha|}{\sin \theta},$

$\rho^2 ||a_2 \times a_3|| = \frac{|\beta|}{\sin \theta},$

(1.28)

thus: $\cos \theta = -\frac{(a_1 \times a_3) \cdot (a_2 \times a_3)}{||a_1 \times a_3|| \, ||a_2 \times a_3||},$

$\alpha = \rho^2 ||a_1 \times a_3|| \sin \theta,$

$\beta = \rho^2 ||a_2 \times a_3|| \sin \theta,$

(1.29)

since the sign of the magnification parameters α and β is normally known in advance and can be taken to be positive.

We can now compute r_1 and r_2 from the second equation in Eq. (1.28) as

$r_1 = \rho^2 \sin \theta$

$$\beta (a_2 \times a_3) = \frac{1}{\|a_2 \times a_3\|}(a_2 \times a_3), \qquad r_2 = r_3 \times r_1. \tag{1.30}$$

Note that there are two possible choices for the matrix R, depending on the value of ε. The translation parameters can now be recovered by writing $Kt = \rho b$, and hence $t = \rho K^{-1} b$. In practical situations, the sign of t_3 is often known in advance (this corresponds to knowing whether the origin of the world coordinate system is in front of or behind the camera), which allows the choice of a unique solution for the calibration parameters.

Figure 1.18 shows the results of an experiment with the dataset from Figure 1.17. The recovered calibration matrix is

$$K = \begin{pmatrix} 970.2841 & 0.0986 & 372.0050 \\ 0 & 963.3466 & 299.2921 \\ 0 & 0 & 1 \end{pmatrix}$$

for this 768 × 576 camera, with estimated values of 1.0072 for the aspect ratio, and 0.0058 degree for the skew angle $|\theta - \pi/2|$.[1] The recovered image center is located about 15 pixels away from the center of the image array.

[1] In this book, an m×n matrix normally has m rows and n columns. Digital images and camera retinas are the only exceptions, and we follow the tradition by assuming that an m × n picture has m columns and n rows. For example, the camera used in this experiment has 768 columns and 576 rows.

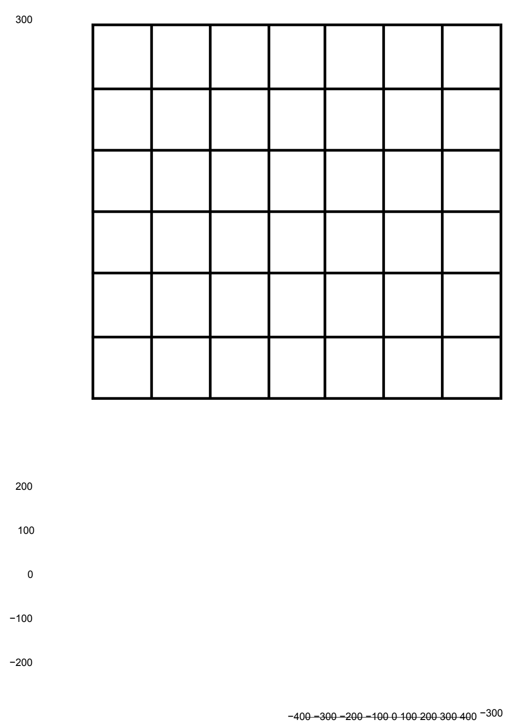


FIGURE 1.18: Results of camera calibration on the dataset shown in Figure 1.17. The original data points (circles) are overlaid with the reprojected 3D points (dots). The root-mean-squared error is 0.96 pixel for this 768 × 576 image.

1.3.2 A Nonlinear Approach to Camera Calibration

The method presented in the previous section ignores some of the constraints as sociated with the calibration process. For example, the camera skew was assumed to be arbitrary instead of (very close to) zero in Section 1.3.1. We present in this section a nonlinear approach to camera calibration that takes into account all the relevant constraints.

This approach is borrowed from photogrammetry, an engineering field whose aim is to recover quantitative geometric information from one or several pictures, with applications in cartography, military intelligence, city planning, etc. For many years, photogrammetry relied on a combination of geometric, optical, and mechani cal methods to recover three-dimensional information from pictures, but the advent of computers in the 1950s has made a purely computational approach to this prob lem feasible. This is the domain of analytical photogrammetry, where the intrinsic parameters of a camera define its interior orientation, and the extrinsic parameters define its exterior orientation.

In this setting, we assume once again that we observe n fiducial points P_i (i =1,...,n) whose positions in some world coordinate system are known, and minimize the mean-squared distance between the measured positions of their im ages and those predicted by the perspective projection equation with respect to a vector of camera parameters ξ in $\mathbb{R}^{11+q}$, where $q \geq 0$, which might include various distortion coefficients in addition to the usual intrinsic and extrinsic parameters. (This assumes that the aspect-ratio and skew are unknown. When they are known, fewer parameters are necessary.) In particular, let us see how to account for radial distortion, a type of aberration that depends on the distance separating the optical axis from the point of interest. We model the projection process by

$$p = \frac{1}{z}\begin{pmatrix} 1/\lambda & 0 & 0 & 0 \\ 0 & 1/\lambda & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \mathcal{M}P, \quad (1.31)$$

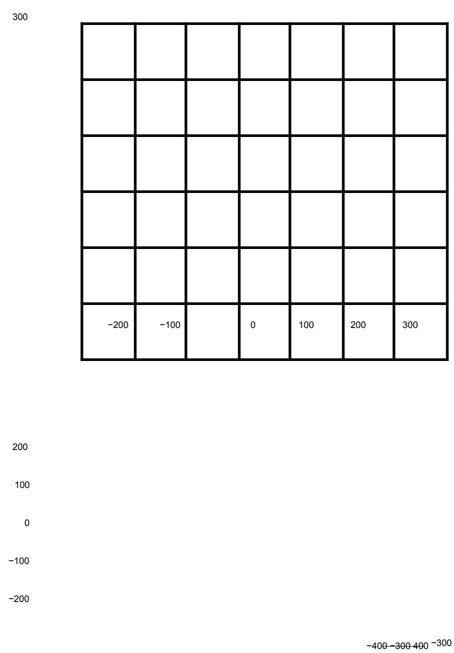


FIGURE 1.19: Results of nonlinear camera calibration on the dataset shown in Figure 1.17. The original data points (circles) are overlaid with the reprojected 3D points (dots). The root-mean-squared error is 0.39 pixel for this 768 × 576 image. Three radial distortion

parameters were used in this case.

where λ is a polynomial function of the squared distance between the image center and the image point p in normalized image coordinates, or:

$$d^2 = \hat{x}^2 + \hat{y}^2 = ||K^{-1}p||^2 - 1. \quad (1.32)$$

In most applications, it is sufficient to use a low-degree polynomial (e.g., $\lambda = 1 + \sum_{p=1}^{q} \kappa_p d^{2p}$, with $q \leq 3$) and the distortion coefficients κ_p ($p = 1,\dots,q$) are normally assumed to be small.

Using Eq. (1.32) to write λ as an explicit function of p in Eq. (1.31) yields highly nonlinear constraints on the 11 + q camera parameters. The least-squares error can be written as

$$E(\xi) = \sum_{i=1}^{2n} f_i^2(\xi), \text{ where } \begin{cases} f_{2j-1}(\xi) = x_j - \dfrac{m_1(\xi) \cdot P_j}{m_3(\xi) \cdot P_j}, \\ f_{2j}(\xi) = y_j - \dfrac{m_2(\xi) \cdot P_j}{m_3(\xi) \cdot P_j}, \end{cases} \text{ for } j = 1, \dots, n. \quad (1.33)$$

Contrary to the cases studied so far, the dependency of each error term $f_i(\xi)$ on the unknown parameters ξ is not linear. Instead, it involves a combination of polynomial and trigonometric functions, and minimizing the overall error measure involves the use of the nonlinear least-squares algorithms discussed in Chapter 22. These algorithms require computing the Jacobian of the vector function $f(\xi) = (f_1[\xi],\dots,f_{2n}[\xi])^T$ with respect to the vector ξ of unknown parameters, which is easily done analytically (see problems).

Figure 1.19 shows the results of an experiment with the dataset from Figure 1.17 using three radial distortion coefficients. The recovered calibration matrix is

$$K = \begin{pmatrix} 1014.0 & 0.0001 & 371.8 \\ 0 & 1008.9 & 292.3 \\ 0 & 0 & 1 \end{pmatrix}$$

Perspective projection

$x = d\dfrac{X}{Z}$

$y = d\dfrac{Y}{Z}$

X, Y, Z: world coordinates (Z< 0)

x, y: image coordinates

d: pinhole-to-retina distance

Weak-perspective projection

$x = -mX$

$y = -mY$

$m = -\dfrac{d}{Z_0}$

X, Y : world coordinates x, y: image coordinates d: pinhole-to-retina distance Z_0: reference-point depth (< 0)

m: magnification (> 0)

projection

Orthographic	$x = X \quad y = Y$	X, Y: world coordinates; x, y: image coordinates
Thin lens equation	$\frac{1}{z} - \frac{1}{Z} = \frac{1}{f}$	z: image-point depth (> 0); Z: object-point depth (< 0); f: focal length

TABLE 1.1: Reference card: Projection models.

for this 768 × 576 camera, with estimated values of 1.0051 for the aspect ratio, and less than 10^{-5} degree for the skew angle. The recovered image center is located about 9 pixels away from the center of the image array. The three radial distortion coefficients are −0.1183, −0.3657, and 1.9112 in this case.

1.4 NOTES

The classical textbook by Hecht (1987) is an excellent introduction to geometric optics, paraxial refraction, thin and thick lenses, and their aberrations, as briefly discussed in Section 1.1. Vignetting is discussed in Horn (1986). Wandell (1995) gives an excellent treatment of image formation in the human visual system. Thor ough presentations of the geometric camera models discussed in Section 1.2 can be found in Faugeras (1993), Hartley and Zisserman (2000b), and Faugeras et al. (2001). The paraperspective projection model was introduced in computer vi sion by Ohta, Maenobu, and Sakai (1981), and its properties have been studied by Aloimonos (1990).

The linear calibration technique described in Section 1.3.1 is detailed in Fau geras (1993). Its variant that takes radial distortion into account is adapted from Tsai (1987). The book of Haralick and Shapiro (1992) presents a concise introduc tion to analytical photogrammetry. The Manual of Photogrammetry is of course the gold standard, and newcomers to this field (like the authors of this book) will prob ably find the ingenious mechanisms and rigorous methods described in the various editions of this book fascinating (Thompson et al. 1966; Slama et al. 1980). We will come back to photogrammetry in the context of structure from motion in Chap ter 8. The linear and nonlinear least-squares techniques used in the approaches to camera calibration discussed in the present chapter are presented in some detail in Chapter 22. An excellent survey and discussion of these methods in the context of analytical photogrammetry can be found in Triggs et al. (2000).

We have assumed in this chapter that a 3D calibration rig is available. This is

equation (homogeneous) $p = \frac{1}{Z} MP$
Perspective projection

Matrix of intrinsic parameters $K =$ θy_0 00 1

α $-\alpha \cot \theta x_0$ 0 $\beta/\sin$

matrix $M = KR\ t$ Perspective projection

Affine projection equation (nonhomogeneous) $p = M$ P 1 $= AP + b$

projection matrix $M = A\ b$ 01

$= \frac{1}{Z_r}$ $R_2\ t_2$

ks

Weak-perspective

TABLE 1.2: Reference card: Geometric camera models.

the setting used in (Faig 1975; Tsai 1987; Faugeras 1993; Heikkil¨a 2000) for exam ple. However, it is difficult to build such a rig accurately—see Lavest, Viala, and Dhome (1998) for a discussion of this problem and an ingenious solution—and many authors prefer using multiple checkerboards or similar planar patterns (Devy, Gar ric & Orteu 1997; Zhang 2000). This includes the widely used C implementation of J.-Y. Bouguet's algorithm, distributed as part of OpenCV, an open-source library of computer vision routines, available at http://opencv.willowgarage.com/wiki/. A MATLAB version is also freely available at his web site; see: http://www. vision.caltech.edu/bouguetj/calib_doc/.

Given the fundamental importance of the notions introduced in this chapter, the main equations derived in its course have been collected in Tables 1.1 and 1.2 for reference.

PROBLEMS

1.1. Derive the perspective equation projections for a virtual image located at a distance d in front of the pinhole.

1.2. Prove geometrically that the projections of two parallel lines lying in some plane Φ appear to converge on a horizon line h formed by the intersection of the image plane Π with the plane parallel to Φ and passing through the pinhole.

1.3. Prove the same result algebraically using the perspective projection Eq. (1.1). You can assume for simplicity that the plane Φ is orthogonal to the image plane Π.

1.4. Consider a camera equipped with a thin lens, with its image plane at position z and the plane of scene points in focus at position Z. Now suppose that the image plane is moved to $\hat{z}$. Show that the diameter of the corresponding blur circle is

$$\frac{a|z - \hat{z}|}{z},$$

where a is the lens diameter. Use this result to show that the depth of field (i.e., the distance between the near and far planes that will keep the diameter

of the blur circles below some threshold ε) is given by

$$D = 2\varepsilon f Z(Z + f)\frac{a}{f^2a^2 - \varepsilon^2 Z^2},$$

and conclude that, for a fixed focal length, the depth of field increases as the lens diameter decreases, and thus the f number f /a increases.

Hint: Solve for the depth $\hat{Z}$ of a point whose image is focused on the image plane at position $\hat{z}$, considering both the case where $\hat{z}$ is larger than z and the case where it is smaller.

1.5. Give a geometric construction of the image p of a point P given the two focal points F and F of a thin lens.

1.6. Show that when the camera coordinate system is skewed and the angle θ be tween the two image axes is not equal to 90 degrees, then Eq. (1.12) transforms into Eq. (1.13).

1.7. Consider two Euclidean coordinate systems $(A)=(O_A, i_A, j_A, k_A)$ and $(B)= (O_B, i_B, j_B, k_B)$. Show that Eq. (1.8) follows from Eq. (1.6).
Hint: First consider the case where both coordinate systems share the same basis vectors—that is, $i_A = i_B$, $j_A = j_B$, and $k_A = k_B$. Then consider the case where they have the same origin—that is, $O_A = O_B$, before finally treating the general case.

1.8. Write formulas for the matrices R in Eq. (1.8) when (B) is deduced from (A) via a rotation of angle θ about the axes i_A, j_A, and k_A respectively. 1.9. Let O denote the homogeneous coordinate vector of the optical center of a cam era in some reference frame, and let M denote the corresponding perspective projection matrix. Show that MO = 0. Explain why this intuitively makes sense.

1.10. Show that the conditions of Theorem 1 are necessary.

1.11. Show that any affine projection matrix M = A b can be written as a general weak-perspective projection matrix as defined by Eq. (1.22), i.e.,

$$M = {}^{1}Z_r \quad ks\ 01 \qquad R_2\, t_2\ .$$

1.12. Give an analytical expression for the Jacobian of the vector function $f(\xi)= (f_1[\xi],...,f_{2n}[\xi])^T$ featured in Eq. (1.33) with respect to the vector ξ of un known parameters.

PROGRAMMING EXERCISES

1.13. Implement the linear calibration algorithm presented in Section 1.3.1.
1.14. Implement the nonlinear calibration algorithm from Section 1.3.2.

CHAPTER 2

Light and Shading

The brightness of a pixel in the image is a function of the brightness of the surface patch in the scene that projects to the pixel. In turn, the brightness of the patch depends on how much incident light arrives at the patch and on the fraction of the incident light that gets reflected (Models in Section 2.1).

This means that the brightness of a pixel is profoundly ambiguous. Surpris ingly, people can disentangle these effects quite accurately. Often, but not always, people can tell whether objects are in bright light or in shadow, and do not perceive objects in shadow as having dark surfaces. People can usually tell whether changes of brightness are caused by changes in reflection or by shading (cinemas wouldn't work if we got it right all the time, however). Typically, people can tell that shading comes from the geometry of the object, but sometimes get shading and markings mixed up. For example, a streak of dark makeup under a cheekbone will often look like a shading effect, making the face look thinner. Quite simple models of shading (Section 2.1) support a range of inference procedures (Section 2.2). More complex models are needed to explain some important effects (Section 2.1.4), but make inference very difficult indeed (Section 2.4).

2.1 MODELLING PIXEL BRIGHTNESS

Three major phenomena determine the brightness of a pixel: the response of the camera to light, the fraction of light reflected from the surface to the camera, and the amount of light falling on the surface. Each can be dealt with quite straight forwardly.

Camera response: Modern cameras respond linearly to middling intensities of light, but have pronounced nonlinearities for darker and brighter illumination. This allows the camera to reproduce the very wide dynamic range of natural light without saturating. For most purposes, it is enough to assume that the camera response is linearly related to the intensity of the surface patch. Write X for a point in space that projects to x in the image, $I_{patch}(X)$ for the intensity of the surface patch at X, and $I_{camera}(x)$ for the camera response at x. Then our model is:

$$I_{camera}(x)= kI_{patch}(x),$$

where k is some constant to be determined by calibration. Generally, we assume that this model applies and that k is known if needed. Under some circumstances, a more complex model is appropriate; we discuss how to recover such models in Section 2.2.1.

Surface reflection: Different points on a surface may reflect more or less of the light that is arriving. Darker surfaces reflect less light, and lighter surfaces reflect more. There is a rich set of possible physical effects, but most can be ignored. Section 2.1.1 describes the relatively simple model that is sufficient for almost all

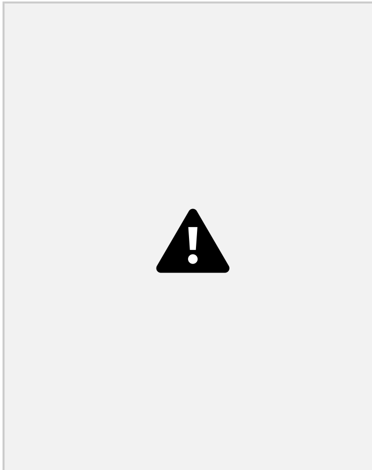


FIGURE 2.1: The two most important reflection modes for computer vision are diffuse re flection (left), where incident light is spread evenly over the whole hemisphere of outgoing directions, and specular reflection (right), where reflected light is concentrated in a single direction. The specular direction S is coplanar with the normal and the source direction (L), and has the same angle to the normal that the source direction does. Most sur faces display both diffuse and specular reflection components. In most cases, the specular component is not precisely mirror like, but is concentrated around a range of directions close to the specular direction (lower right). This causes specularities, where one sees a mirror like reflection of the light source. Specularities, when they occur, tend to be small and bright. In the photograph, they appear on the metal spoon and on the plate. Large specularities can appear on flat metal surfaces (arrows). Most curved surfaces (such as the plate) show smaller specularities. Most of the reflection here is diffuse; some cases are indicated by arrows. Martin Brigdale c Dorling Kindersley, used with permission.

purposes in computer vision.

Illumination: The amount of light a patch receives depends on the overall intensity of the light, and on the geometry. The overall intensity could change because some luminaires (the formal term for light sources) might be shadowed, or might have strong directional components. Geometry affects the amount of light arriving at a patch because surface patches facing the light collect more radiation and so are brighter than surface patches tilted away from the light, an effect known as shading. Section 2.1.2 describes the most important model used in computer vision; Section 2.3 describes a much more complex model that is necessary to explain some important practical difficulties in shading inference.

2.1.1 Reflection at Surfaces

Most surfaces reflect light by a process of diffuse reflection. Diffuse reflection scat ters light evenly across the directions leaving a surface, so the brightness of a diffuse surface doesn't depend on the viewing direction. Examples are easy to identify with this test: most cloth has this property, as do most paints, rough wooden surfaces, most vegetation, and rough stone or concrete. The only parameter required to describe a surface of this type is its albedo, the fraction of the light arriving at the surface that is reflected. This does not depend on the direction in which the light arrives or the direction in which the light leaves. Surfaces with very high or very low albedo are difficult to make. For practical surfaces, albedo lies in the range 0.05 – 0.90 (see Brelstaff and Blake (1988b), who argue the dynamic range is closer to 10 than the 18 implied by these numbers). Mirrors are not diffuse, because what you see depends on the direction in which you look at the mirror. The behav ior of a perfect mirror is known as specular reflection. For an ideal mirror, light arriving along a particular direction can leave only along the specular direction, obtained by reflecting the direction of incoming radiation about the surface normal (Figure 2.1). Usually some fraction of incoming radiation is absorbed; on an ideal specular surface, this fraction does not depend on the incident direction.

If a surface behaves like an ideal specular reflector, you could use it as a mirror, and based on this test, relatively few surfaces actually behave like ideal specular reflectors. Imagine a near perfect mirror made of polished metal; if this surface suffers slight damage at a small scale, then around each point there will be a set of small facets, pointing in a range of directions. In turn, this means that light arriving in one direction will leave in several different directions because it strikes several facets, and so the specular reflections will be blurred. As the surface becomes less flat, these distortions will become more pronounced; eventually, the only specular reflection that is bright enough to see will come from the light source. This mechanism means that, in most shiny paint, plastic, wet, or brushed metal surfaces, one sees a bright blob—often called a specularity—along the specular di rection from light sources, but few other specular effects. Specularities are easy to identify, because they are small and very bright (Figure 2.1; Brelstaff and Blake (1988b)). Most surfaces reflect only some of the incoming light in a specular compo nent, and we can represent the percentage of light that is specularly reflected with a specular albedo. Although the diffuse albedo is an important material property that we will try to estimate from images, the specular albedo is largely seen as a nuisance and usually is not estimated.

2.1.2 Sources and Their Effects

The main source of illumination outdoors is the sun, whose rays all travel parallel to one another in a known direction because it is so far away. We model this behavior with a distant point light source. This is the most important model of lighting (because it is like the sun and because it is easy to use), and can be quite effective for indoor scenes as well as outdoor scenes. Because the rays are parallel to one another, a surface that faces the source cuts more rays (and so collects more light) than one oriented along the direction in which the rays travel. The amount of light collected by a surface patch in this model is proportional to the cosine of the angle θ between the illumination direction and the normal (Figure 2.2). The figure yields Lambert's cosine law, which states the brightness of a diffuse patch illuminated by a distant point light source is given by

$$I = \rho I_0 \cos \theta,$$

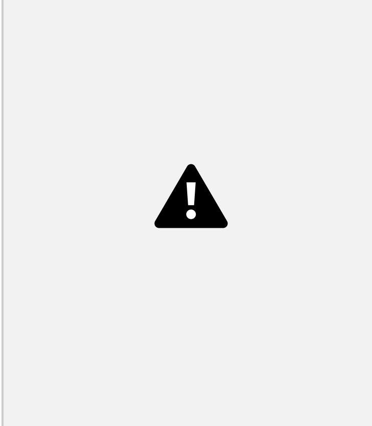


FIGURE 2.2: The orientation of a surface patch with respect to the light affects how much light the patch gathers. We model surface patches as illuminated by a distant point source, whose rays are shown as light arrowheads. Patch A is tilted away from the source (θ is close to 90^0) and collects less energy, because it cuts fewer light rays per unit surface area. Patch B, facing the source (θ is close to 0^0), collects more energy, and so is brighter. Shadows occur when a patch cannot see a source. The shadows are not dead black, because the surface can see interreflected light from other surfaces. These effects are shown in the photograph. The darker surfaces are turned away from the illumination direction. Martin Brigdale c Dorling Kindersley, used with permission.

where I_0 is the intensity of the light source, θ is the angle between the light source direction and the surface normal, and ρ is the diffuse albedo. This law predicts that bright image pixels come from surface patches that face the light directly and dark pixels come from patches that see the light only tangentially, so that the shading on a surface provides some shape information. We explore this cue in Section 2.4.

If the surface cannot see the source, then it is in shadow. Since we assume that light arrives at our patch only from the distant point light source, our model suggests that shadows are deep black; in practice, they very seldom are, because the shadowed surface usually receives light from other sources. Outdoors, the most important such source is the sky, which is quite bright. Indoors, light reflected from other surfaces illuminates shadowed patches. This means that, for example, we tend to see few shadows in rooms with white walls, because any shadowed patch receives a lot of light from the walls. These interreflections also can have a significant effect on the brightness surfaces that are not in shadow. Interreflection effects are sometimes modelled by adding a constant ambient illumination term to the predicted intensity. The ambient term ensures that shadows are not too dark, but this is not a particularly successful model of the spatial properties of interreflections. More detailed models require some familiarity with radiometric terminology, but they are important in some applications; we have confined this topic to Section 2.3.

2.1.3 The Lambertian+Specular Model

For almost all purposes, it is enough to model all surfaces as being diffuse with specularities. This is the lambertian+specular model. Specularities are relatively seldom used in inference (Section 2.2.2 sketches two methods), and so there is no need for a formal model of their structure. Because specularities are small and bright, they are relatively easy to identify and remove with straightforward methods (find small bright spots, and replace them by smoothing the local pixel values). More sophisticated specularity finders use color information (Section 3.5.1). Thus, to apply the lambertian+specular model, we find and remove specularities, and then use Lambert's law (Section 2.1.2) to model image intensity.

We must choose which source effects to model. In the simplest case, a local shading model, we assume that shading is caused only by light that comes from the luminaire (i.e., that there are no interreflections).

Now assume that the luminaire is an infinitely distant source. For this case, write $N(x)$ for the unit surface normal at x, S for a vector pointing from x toward the source with length I_o (the source intensity), $\rho(x)$ for the albedo at x, and $Vis(S, x)$ for a function that is 1 when x can see the source and zero otherwise. Then, the intensity at x is

$$I(x)= \rho(x)(N \cdot S)\, Vis(S, x)+ \rho(x)A + M$$

Image = Diffuse + Ambient + Specular (mirror-like) intensity term term term

This model can still be used for a more complex source (for example, an area source), but in that case it is more difficult to determine an appropriate $S(x)$.

2.1.4 Area Sources

An area source is an area that radiates light. Area sources occur quite com monly in natural scenes—an overcast sky is a good example—and in synthetic environments—for example, the fluorescent light boxes found in many industrial ceilings. Area sources are common in illumination engineering, because they tend not to cast strong shadows and because the illumination due to the source does not fall off significantly as a function of the distance to the source. Detailed models of area sources are complex (Section 2.3), but a simple model is useful to understand shadows. Shadows from area sources are very different from shadows cast by point sources. One seldom sees dark shadows with crisp boundaries indoors. Instead, one could see no visible shadows, or shadows that are rather fuzzy diffuse blobs, or sometimes fuzzy blobs with a dark core (Figure 2.3). These effects occur because rooms tend to have light walls and diffuse ceiling fixtures, which act as area sources. As a result, the shadows one sees are area source shadows.

To compute the intensity at a surface patch illuminated by an area source, we can break the source up into infinitesimal source elements, then sum effects from each element. If there is an occluder, then some surface patches may see none of the source elements. Such patches will be dark, and lie in the umbra (a Latin word meaning "shadow"). Other surface patches may see some, but not all, of the source elements. Such patches may be quite bright (if they see most of the elements), or

FIGURE 2.3: Area sources generate complex shadows with smooth boundaries, because from the point of view of a surface patch, the source disappears slowly behind the occluder. Left: a photograph, showing characteristic area source shadow effects. Notice that A is much darker than B; there must be some shadowing effect here, but there is no clear shadow boundary. Instead, there is a fairly smooth gradient. The chair leg casts a complex shadow, with two distinct regions. There is a core of darkness (the umbra—where the source cannot be seen at all) surrounded by a partial shadow (penumbra—where the source can be seen partially). A good model of the geometry, illustrated right, is to imagine lying with your back to the surface looking at the world above. At point 1, you can see all of the source; at point 2, you can see some of it; and at point 3, you can see none of it. Peter Anderson c Dorling Kindersley, used with permission.

relatively dark (if they see few elements), and lie in the penumbra (a compound of Latin words meaning “almost shadow”). One way to build intuition is to think of a tiny observer looking up from the surface patch. At umbral points, this observer will not see the area source at all whereas at penumbral points, the observer will see some, but not all, of the area source. An observer moving from outside the shadow, through the penumbra and into the umbra will see something that looks like an eclipse of the moon (Figure 2.3). The penumbra can be large, and can change quite slowly from light to dark. There might even be no umbral points at all, and, if the occluder is sufficiently far away from the surface, the penumbra could be very large and almost indistinguishable in brightness from the unshadowed patches. This is why many objects in rooms appear to cast no shadow at all (Figure 2.4).

2.2 INFERENCE FROM SHADING

Shading can be used to infer a variety of properties of the visual world. Successful inference often requires that we calibrated the camera radiometrically, so that we know how pixel values map to radiometric values (Section 2.2.1). As Figure 2.1

FIGURE 2.4: The photograph on the left shows a room interior. Notice the lighting has some directional component (the vertical face indicated by the arrow is dark, because it does not face the main direction of lighting), but there are few visible shadows (for example, the chairs do not cast a shadow on the floor). On the right, a drawing to show why; here there is a small occluder and a large area source. The occluder is some way away from the shaded surface. Generally, at points on the shaded surface the incoming hemisphere looks like that at point 1. The occluder blocks out some small percentage of the area source, but the amount of light lost is too small to notice (compare figure 2.3). Jake Fitzjones c Dorling Kindersley, used with permission.

suggests, specularities are a source of information about the shape of a surface, and Section 2.2.2 shows how this information can be interpreted. Section 2.2.3 shows how to recover the albedoes of surfaces from images. Finally, Section 2.2.4 shows how multiple shaded images can be used to recover surface shape.

2.2.1 Radiometric Calibration and High Dynamic Range Images

Real scenes often display a much larger range of intensities than cameras can cope with. Film and charge-coupled devices respond to energy. A property called reci procity means that, if a scene patch casts intensity E onto the film, and if the shutter is open for time Δt, the response is a function of $E\Delta t$ alone. In particular, we will get the same outcome if we image one patch of intensity E for time Δt and another patch of intensity E/k for time $k\Delta t$. The actual response that the film produces is a function of $E\Delta t$; this function might depend on the imaging system, but is typically somewhat linear over some range, and sharply non-linear near the top and bottom of this range, so that the image can capture very dark and very light patches without saturation. It is usually monotonically increasing.

There are a variety of applications where it would be useful to know the actual radiance (equivalently, the intensity) arriving at the imaging device. For example, we might want to compare renderings of a scene with pictures of the scene, and to do that we need to work in real radiometric units. We might want to use pictures of a scene to estimate the lighting in that scene so we can postrender new objects into the scene, which would need to be lit correctly. To infer radiance, we must determine the film response, a procedure known as radiometric calibration. As we

shall see, doing this will require more than one image of a scene, each obtained at different exposure settings. Imagine we are looking at a scene of a stained glass window lit from behind in a church. At one exposure setting, we would be able to resolve detail in the dark corners, but not on the stained glass, which would be saturated. At another setting, we would be able to resolve detail on the glass, but the interior would be too dark. If we have both settings, we may as well try to recover radiance with a very large dynamic range—producing a high dynamic range image.

Now assume we have multiple registered images, each obtained using a different exposure time. At the i, j'th pixel, we know the image intensity value $I^{(k)}_{ij}$ for the k'th exposure time, we know the value of the k'th exposure time Δt_k, and we know that the intensity of the corresponding surface patch E_{ij} is the same for each exposure, but we do not know the value of E_{ij} . Write the camera response function f, so that

$$I^{(k)}_{ij} = f(E_{ij}\Delta t_k).$$

There are now several possible approaches to solve for f. We could assume a parametric form—say, polynomial—then solve using least squares. Notice that we must solve not only for the parameters of f, but also for E_{ij} . For a color camera, we solve for calibration of each channel separately. Mitsunaga and Nayar (1999) have studied the polynomial case in detail. Though the solution is not unique, ambiguous solutions are strongly different from one another, and most cases are easily ruled out. Furthermore, one does not need to know exposure times with exact accuracy to estimate a solution, as long as there are sufficient pixel values; instead, one estimates f from a fixed set of exposure times, then estimates the exposure times from f, and then re-estimates. This procedure is stable.

Alternatively, because the camera response is monotonic, we can work with its inverse $g = f^{-1}$, take logs, and write

$$\log g(I^{(k)}_{ij}) = \log E_{ij} + \log \Delta t_k.$$

We can now estimate the values that g takes at each point and the E_{ij} by placing a smoothness penalty on g. In particular, we minimize

$$\sum_{i,j,k} (\log g(I^{(k)}_{ij}) - (\log E_{ij} + \log \Delta t_k))^2 + \text{smoothness penalty on } g$$

by choice of g. Debevec and Malik (1997) penalize the second derivative of g. Once we have a radiometrically calibrated camera, estimating a high dynamic range image is relatively straightforward. We have a set of registered images, and at each pixel location, we seek the estimate of radiance that predicts the registered image values best. In particular, we assume we know f. We seek an E_{ij} such that

$$\sum_{k} w(I_{ij})(I^{(k)}_{ij} - f(E_{ij}\Delta t_k))^2$$

is minimized. Notice the weights because our estimate of f is more reliable when I_{ij} is in the middle of the available range of values than when it is at larger or smaller values.

FIGURE 2.5: It is possible to calibrate the radiometric response of a camera from multiple images obtained at different exposures. The top row shows four different exposures of the same scene, ranging from darker (shorter shutter time) to lighter (longer shutter time). Note how, in the dark frames, the lighter part of the image shows detail, and in the light frames, the darker part of the image shows detail; this is the result of non-linearities in the camera response. On the bottom left, we show the inferred calibration curves for each of the R, G, and B camera channels. On the bottom right, a composite image illustrates the results. The dynamic range of this image is far too large to print; instead, the main image is normalized to the print range. Overlaid on this image are boxes where the radiances in the box have also been normalized to the print range; these show how much information is packed into the high dynamic range image. This figure was originally published as Figure 7 of “Radiometric Self Calibration,” by T. Mitsunaga and S. Nayar, Proc. IEEE CVPR 1999, c IEEE, 1999.

2.2.2 The Shape of Specularities

Specularities are informative. They offer hints about the color of illumination (see Chapter 3) and offer cues to the local geometry of a surface. Understanding these cues is a useful exercise in differential geometry. We consider a smooth specular surface and assume that the brightness reflected in the direction V is a function of $V \cdot P$, where P is the specular direction. We expect the specularity to be small and isolated, so we can assume that the source direction S and the viewing direction V are constant over its extent. Let us further assume that the specularity can be defined by a threshold on the specular energy, i.e., $V \cdot P \geq 1 - \varepsilon$ for some constant ε, denote by N the unit surface normal, and define the half-angle direction as $H = (S + V)/2$ (Figure 2.6(left)). Using the fact that the vectors S, V and P have unit length and a whit of plane geometry, it can easily be shown that the boundary of the specularity is defined by (see exercises)

$$1 - \varepsilon = V \cdot P = 2\frac{(H \cdot N)^2}{(H \cdot H)} - 1. \quad (2.1)$$

Because the specularity is small, the second-order structure of the surface

$\alpha/2$ *z*

P
H N

S
H N

V

α

u u
2
1

x y

FIGURE 2.6: A specular surface viewed by a distant observer. We establish a coordinate system at the brightest point of the specularity (where the half-angle direction is equal to the normal) and orient the system using the normal and principal directions.

will allow us to characterize the shape of its boundary as follows: there is some point on the surface inside the specularity (in fact, the brightest point) where H is parallel to N. We set up a coordinate system at this point, oriented so that the z-axis lies along N and the x- and y-axes lie along the principal directions u_1 and u_2 (Figure 2.6(right)). As noted earlier, the surface can be represented up to second order as $z = -1/2(\kappa_1 x^2 + \kappa_2 y^2)$ in this frame, where κ_1 and κ_2 are the principal curvatures. Now, let us define a parametric surface as a differentiable mapping $x : U \subset R^2 \rightarrow R^3$ associating with any couple $(u, v) \in U$ the coordinate vector $(x, y, z)^T$ of a point in some fixed coordinate system. It is easily shown (see exercises) that the normal to a parametric surface is along the vector $\frac{\partial}{\partial u} x \times \frac{\partial}{\partial v} x$. Our second-order surface model is a parametric surface parameterized by x and y, thus its unit surface normal is defined in the corresponding frame by

$$N(x, y) = \frac{1}{\sqrt{1 + \kappa_1^2 x^2 + \kappa_2^2 y^2}} \begin{pmatrix} \kappa_1 x \\ \kappa_2 y \\ 1 \end{pmatrix},$$

and $H = (0, 0, 1)^T$. Because H is a constant, we can rewrite Eq. (2.1) as $\kappa_1^2 x^2 + \kappa_2^2 y^2 = \zeta$, where ζ is a constant depending on ε. In particular, the shape of the specularity on the surface contains information about the second fundamental form. The specularity will be an ellipse, with major and minor axes oriented along the principal directions, and an eccentricity equal to the ratio of the principal curvatures. Unfortunately, the shape of the specularity on the surface is not, in general, directly observable, so this property can be exploited only when a fair amount about the viewing and illumination setup is known (Healey and Binford 1986).

Although we cannot get much out of the shape of the specularity in the image, it is possible to tell a convex surface from a concave one by watching how a

specularity moves as the view changes (you can convince yourself of this with the aid of a spoon).[1] Let us consider a point source at infinity and assume that the specular lobe is very narrow, so the viewing direction and the specular direction coincide. Initially, the specular direction is V , and the specularity is at the surface point P; after a small eye motion, V changes to V , while the specularity moves to the close-by point P (Figure 2.7).

Motion of observer

Motion of observer

N' V N

Source direction

V'

V N N'

Motion of V'

P P'

P' P

specularity Motion of specularity

FIGURE 2.7: Specularities on convex and concave surfaces behave differently when the view changes. With an appropriate choice of source direction and motion, this could be used to obtain the signs of the principal curvatures.

The quantity of interest is $\delta a = (V - V) \cdot t$, where $t = \frac{1}{\delta s}\overrightarrow{PP}$ is tangent to the surface, and δs is the (small) distance between P and P : if δa is positive, then the specularity moves in the direction of the view (back of the spoon), and if it is negative, the specularity moves against the direction of the view (bowl of the spoon). By construction, we have $V = 2(S \cdot N)N - S$, and

$$\begin{aligned} V &= 2(S \cdot N)N - S = 2(S \cdot (N + \delta N))(N + \delta N) - S \\ &= V + 2(S \cdot \delta N)N + 2(S \cdot N)\delta N + 2(S \cdot \delta N)\delta N, \end{aligned}$$

where $\delta N \stackrel{\text{def}}{=} N - N = \delta s\, dN(t)$. Because t is tangent to the surface in P, ignoring second-order terms yields

$$\delta a = (V - V) \cdot t = 2(S \cdot N)(\delta N \cdot t) = 2(S \cdot N)(\delta s)(\mathrm{II}(t, t)).$$

Thus, for a concave surface, the specularity always moves against the view, and for a convex surface, it always moves with the view. Things are more complex with hyperbolic surfaces; the specularity may move with the view, against the view, or perpendicular to the view (when t is an asymptotic direction).

[1] Of course, there is a simpler way to distinguish (by sight) the concave bowl of a spoon from its convex back: it is just the side where your reflection is upside down. This property of concave mirrors was demonstrated to one of the authors by one of his friends at a dinner party, causing much

consternation since the author in question was at the time bragging about differential geometry, and his friend, who does not pretend to know anything about mathematics, just looked at herself in the spoon.

log ρ

log I log p

dlog ρ
dx dlog I

dx dlog p

dx

dlog p

dx Thresholded

Integrate
this to get
lightness,
which is
log ρ + constant

FIGURE 2.8: The lightness algorithm is easiest to illustrate for a 1D image. In the top row, the graph on the left shows log ρ(x), that in the center log I(x), and that on the right their sum, which is log C. The log of image intensity has large derivatives at changes in surface reflectance and small derivatives when the only change is due to illumination gradients. Lightness is recovered by differentiating the log intensity, thresholding to dis pose of small derivatives, and integrating at the cost of a missing constant of integration.

2.2.3 Inferring Lightness and Illumination

If we could estimate the albedo of a surface from an image, then we would know a property of the surface itself, rather than a property of a picture of the surface.

Such properties are often called intrinsic representations. They are worth estimating, because they do not change when the imaging circumstances change. It might seem

that albedo is difficult to estimate, because there is an ambiguity linking albedo and illumination; for example, a high albedo viewed under middling illumination will give the same brightness as a low albedo viewed under bright light. However, humans can report whether a surface is white, gray, or black (the lightness of the surface), despite changes in the intensity of illumination (the brightness). This skill is known as lightness constancy. There is a lot of evidence that human lightness constancy involves two processes: one process compares the brightness of various image patches and uses this comparison to determine which patches are lighter and which darker; the second establishes some form of absolute standard to which these comparisons can be referred (e.g. Gilchrist et al. (1999)).

Current lightness algorithms were developed in the context of simple scenes. In particular, we assume that the scene is flat and frontal; that surfaces are diffuse, or that specularities have been removed; and that the camera responds linearly. In this case, the camera response C at a point x is the product of an illumination term, an albedo term, and a constant that comes from the camera gain:

$$C(x)= k_c I(x)\rho(x).$$

If we take logarithms, we get

$$\log C(x) = \log k_c + \log I(x) + \log \rho(x).$$

We now make a second set of assumptions:

- First, we assume that albedoes change only quickly over space. This means that a typical set of albedoes will look like a collage of papers of different grays. This assumption is quite easily justified: There are relatively few continuous changes of albedo in the world (the best example occurs in ripening fruit), and changes of albedo often occur when one object occludes another (so we would expect the change to be fast). This means that spatial derivatives of the term log $\rho(x)$ are either zero (where the albedo is constant) or large (at a change of albedo).

- Second, illumination changes only slowly over space. This assumption is somewhat realistic. For example, the illumination due to a point source will change relatively slowly unless the source is very close, so the sun is a par ticularly good source for this method, as long as there are no shadows. As another example, illumination inside rooms tends to change very slowly be cause the white walls of the room act as area sources. This assumption fails dramatically at shadow boundaries, however. We have to see these as a spe cial case and assume that either there are no shadow boundaries or that we know where they are.

We can now build algorithms that use our model. The earliest algorithm is the Retinex algorithm of Land and McCann (1971); this took several forms, most of which have fallen into disuse. The key insight of Retinex is that small gradients are changes in illumination, and large gradients are changes in lightness. We can use this by differentiating the log transform, throwing away small gradients, and integrating the results (Horn 1974); these days, this procedure is widely known as

Retinex. There is a constant of integration missing, so lightness ratios are available, but absolute lightness measurements are not. Figure 2.8 illustrates the process for a one-dimensional example, where differentiation and integration are easy.

This approach can be extended to two dimensions as well. Differentiating and thresholding is easy: at each point, we estimate the magnitude of the gradient; if the magnitude is less than some threshold, we set the gradient vector to zero; otherwise, we leave it alone. The difficulty is in integrating these gradients to get the log albedo map. The thresholded gradients may not be the gradients of an image because the mixed second partials may not be equal (integrability again; compare with Section 2.2.4).

Form the gradient of the log of the image
At each pixel, if the gradient magnitude is below
  a threshold, replace that gradient with zero
Reconstruct the log-albedo by solving the minimization
  problem described in the text
Obtain a constant of integration
Add the constant to the log-albedo, and exponentiate

Algorithm 2.1:
Determining the Lightness of Image Patches.

The problem can be rephrased as a minimization problem: choose the log albedo map whose gradient is most like the thresholded gradient. This is a relatively simple problem because computing the gradient of an image is a linear operation. The x-component of the thresholded gradient is scanned into a vector p, and the y component is scanned into a vector q. We write the vector representing log-albedo as l. Now the process of forming the x derivative is linear, and so there is some matrix M_x, such that $M_x l$ is the x derivative; for the y derivative, we write the corresponding matrix M_y.

The problem becomes finding the vector l that minimizes

$$|M_x l - p|^2 + |M_y l - q|^2 .$$

This is a quadratic minimization problem, and the answer can be found by a linear process. Some special tricks are required because adding a constant vector to l cannot change the derivatives, so the problem does not have a unique solution. We explore the minimization problem in the exercises.

The constant of integration needs to be obtained from some other assumption. There are two obvious possibilities:

- we can assume that the brightest patch is white;
- we can assume that the average lightness is constant.

We explore the consequences of these models in the exercises.

More sophisticated algorithms are now available, but there were no quan titative studies of performance until recently. Grosse et al. built a dataset for

FIGURE 2.9: Retinex remains a strong algorithm for recovering albedo from images. Here we show results from the version of Retinex described in the text applied to an image of a room (left) and an image from a collection of test images due to Grosse et al. (2009). The center-left column shows results from Retinex for this image, and the center right column shows results from a variant of the algorithm that uses color reasoning to improve the classification of edges into albedo versus shading. Finally, the right column shows the correct answer, known by clever experimental methods used when taking the pictures. This problem is very hard; you can see that the albedo images still contain some illumination signal. Part of this figure courtesy Kevin Karsch, U. Illinois. Part of this figure was originally published as Figure 3 of “Ground truth dataset and baseline evaluations for intrinsic image algorithms,” by R. Grosse, M. Johnson, E. Adelson, and W. Freeman, Proc. IEEE ICCV 2009, c IEEE, 2009.

evaluating lightness algorithms, and show that a version of the procedure we de scribe performs extremely well compared to more sophisticated algorithms

(2009). The major difficulty with all these approaches is caused by shadow boundaries, which we discuss in Section 3.5.2.

2.2.4 Photometric Stereo: Shape from Multiple Shaded Images

It is possible to reconstruct a patch of surface from a series of pictures of that surface taken under different illuminants. First, we need a camera model. For simplicity, we choose a camera situated so that the point (x, y, z) in space is imaged to the point (x, y) in the camera (the method we describe works for the other camera models described in Chapter 1).

In this case, to measure the shape of the surface, we need to obtain the depth to the surface. This suggests representing the surface as (x, y, f(x, y))—a

representation known as a Monge patch after the French military engineer who first used it (Figure 2.10). This representation is attractive because we can determine a unique point on the surface by giving the image coordinates. Notice that to obtain a measurement of a solid object, we would need to reconstruct more than one patch because we need to observe the back of the object.

Image
plane

direction
of projection

height

y

x

FIGURE 2.10: A Monge patch is a representation of a piece of surface as a height function. For the photometric stereo example, we assume that an orthographic camera—one that maps (x, y, z) in space to (x, y) in the camera—is viewing a Monge patch. This means that the shape of the surface can be represented as a function of position in the image.

Photometric stereo is a method for recovering a representation of the Monge patch from image data. The method involves reasoning about the image intensity values for several different images of a surface in a fixed view illuminated by different sources. This method recovers the height of the surface at points corresponding to each pixel; in computer vision circles, the resulting representation is often known as a height map, depth map, or dense depth map.

Fix the camera and the surface in position, and illuminate the surface using a point source that is far away compared with the size of the surface. We adopt a local shading model and assume that there is no ambient illumination (more about this later) so that the brightness at a point x on the surface is

$$B(x)= \rho(x)N(x) \cdot S_1,$$

where N is the unit surface normal and S_1 is the source vector. We can write B(x, y) for the radiosity of a point on the surface because there is only one point on the surface corresponding to the point (x, y) in the camera. Now we assume that the response of the camera is linear in the surface radiosity, and so have that the

value of a pixel at (x, y) is

$$
\begin{aligned}
I(x, y) &= kB(x) \\
&= kB(x, y) \\
&= k\rho(x, y)N(x, y) \cdot S_1 \\
&= g(x, y) \cdot V_1,
\end{aligned}
$$

where $g(x, y) = \rho(x, y)N(x, y)$ and $V_1 = kS_1$, where k is the constant connecting the camera response to the input radiance.

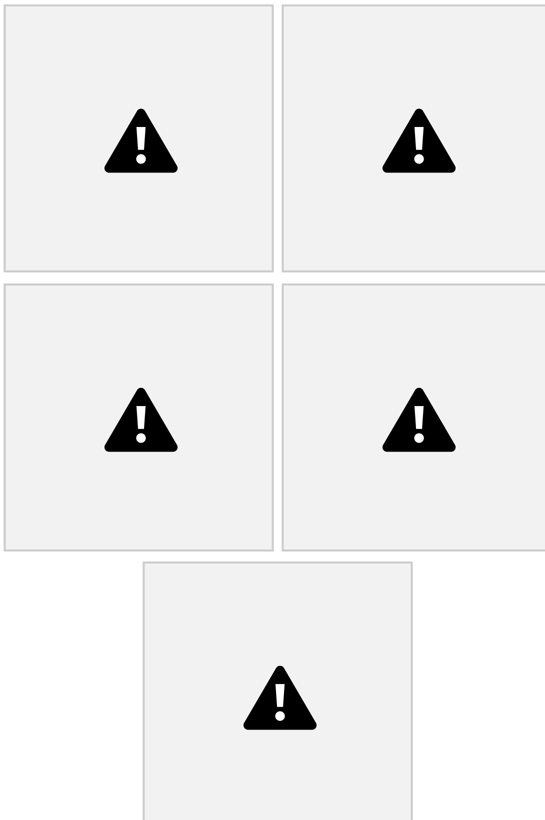


FIGURE 2.11: Five synthetic images of a sphere, all obtained in an orthographic view from the same viewing position. These images are shaded using a local shading model and a distant point source. This is a convex object, so the only view where there is no visible shadow occurs when the source direction is parallel to the viewing direction. The variations in brightness occuring under different sources code the shape of the surface.

In these equations, $g(x, y)$ describes the surface, and V_1 is a property of the illumination and of the camera. We have a dot product between a vector field $g(x, y)$ and a vector V_1, which could be measured; with enough of these dot products, we could reconstruct g and so the surface.

Now if we have n sources, for each of which V_i is known, we stack each of these V_i into a known matrix V, where

$$V = \left(V^T_1 \quad V^T_2 \right.$$

$$\vdots \quad V^T{}_n \quad) \quad ||) .$$

For each image point, we stack the measurements into a
vector $i(x, y)= \{\mathbf{I}_1(x, y),\mathbf{I}_2(x, y),...,\mathbf{I}_n(x, y)\}^T$.